Supplementary Information

Antisolvent crystallization for nickel sulfate production: an 8000 t/y plant design and resource assessment

This supplement contains model inputs, property data, resource ledgers, numerical verification and secondary comparisons. Tables retaining O/C allocations record endpoint heat placement; Table S34 distinguishes composition-specific and preheat targets. Main-text figures and tables use unprefixed numbers; supplementary items use S numbers.

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S1 Crystallization data basis and thermodynamic inputs

Table S1 preserves the denominator of representative Demirel et al. nickel measurements. The reported liquid nickel amount is expressed per initial aqueous volume, while the present process endpoint is expressed per final mother-liquor mass. The two values describe related but different compositions and are kept on separate bases.

Table S1: Selected 25°C pure-salt acetone-series data from Demirel et al. [1], Table 5. The initial solution contained 1.00 mol/L Ni, had pH 0.5 and 132.5 g/L total sulfate. Nickel values retain the published initial-aqueous-volume denominator; O/A is nominal added organic/initial aqueous volume. Mass values convert the paper's rounded molar values and do not add precision.

Nominal O/A	Printed Ni, mol/L	Converted Ni, g/L	Interpretation
0.5	0.32	18.8	Conversion of rounded value
1.0	0.11	6.5	Conversion of rounded value
2.0	0.02	1.17	Text reports about 1.3 g/L

The 1.17 g/L entry in the last row is 0.02×58.693 applied to a printed two-decimal molar concentration; the article separately describes the result as approximately 1.3 g/L. The time-to-equilibrium study was performed at O/A=1 and should not be read as a separate 24 h confirmation of every point in Table S1. The post-ion-exchange liquor is a separate experimental matrix: its initial nickel concentration was 4.544 g/L and it contained Al, Co, Fe, Mg, Mn, Na and Zn, while free-acid and total-sulfate concentrations were not specified. Demirel et al.'s 3 h pure-acetone experiment B reported 1.681 g/L Ni per initial aqueous volume and 62.10% recovery; the staged diluted-acetone experiment E reported 1.505 g/L and 54.06%, respectively. These are operating endpoints rather than the pure-salt equilibrium points in Table S1. The process scenarios specify 0.700 or 0.820 g/kg Ni in the *final* liquid and are compared through full-plant balances on that basis.

S1.1 Acetone–water binary parameters

Both sets use $\tau_{ij} = a_{ij} + b_{ij}/T + e_{ij} \ln(T/1\,\mathrm{K})$, with absolute temperature T in kelvin. Component 1 is acetone and component 2 is water. The nonrandomness parameter is constant, $\alpha_{12} = \alpha_{21} = 0.3$; the linear-in-$T$ coefficients are zero. The temperature bounds stored with a parameter entry specify its model range, while the experimental checks describe the evidence supporting its use.

Table S2: Acetone–water NRTL parameters. Dimensionless coefficients use the kelvin temperature convention stated above. Stored bounds are shown in Celsius.

Parameter	Original	Alternative
a_{12}	−3.076800000	−70.690506983
a_{21}	7.938500000	81.462249306
b_{12} / K	1203.729200	4120.536877
b_{21} / K	−2099.673600	−4758.873114
e_{12}	0	10.148990081
e_{21}	0	−11.276536115
$\alpha_{12} = \alpha_{21}$	0.3	0.3
Stored temperature bounds / °C	20–230	20–87.45

The alternative regression uses 40 VLE points and 52 mixing-enthalpy points over 293.15–

360.60 K. Its least-squares objective is

$$\Phi = \frac{1}{2n_V}\sum_{k=1}^{n_V}\sum_{i=1}^{2}\left(\frac{\ln\gamma_{ik}^{\mathrm{model}} - \ln\gamma_{ik}^{\mathrm{target}}}{0.02}\right)^2 + \frac{1}{n_H}\sum_{k=1}^{n_H}\left(\frac{H_k^{E,\mathrm{model}} - H_k^{E,\mathrm{data}}}{50\ \mathrm{J\,mol^{-1}}}\right)^2, \tag{S1}$$

where the activity-coefficient targets are derived from the VLE data, $n_V = 40$, and $n_H = 52$. The normalization gives the two data types equal total weight and points within each type equal weight. The scales 0.02 and 50 J/mol balance VLE and calorimetry; they are not measurement standard deviations. The other 32 VLE and 39 enthalpy points are withheld from fitting, including calorimetry at 283.15 K. Their original-model residuals had previously been inspected, so the holdout is source-separated rather than a blind new measurement set.

The archive retains the full-precision coefficients, Aspen-entry mapping, original parameter records and regression objective. Calculations above 87.45°C, including the approximately 101°C BASE95-ALT reboiler, extrapolate the alternative set. The sensitivity changes only acetone–water interactions.

S1.2 Saved dielectric-parameter record

The saved dielectric correlation is $D = A + B(1/T - 1/C)$, with T and C in kelvin, dimensionless A and B in kelvin. Database-cache values and the effective saved summary are distinct records. Table S3 compares them for the W700 backup and exported report. The global ENRTL–RK/GLOBAL framework also appears in the four audited HeatX-side readbacks. The GMENRTLQ substitution warning occurs during global property initialization; the log does not resolve its effect by unit operation.

Table S3: Initial dielectric (CPDIEC) parameter records for saved W700. The explicit-input comparison below tests the acid coefficient choice separately.

Component	Cached (A, B, C)	Effective saved record
Acetone	$(20.82882, 7831.274, 298.15)$	Present; report agrees
Water	$(78.54003, 31989.38, 298.15)$	Present; report agrees
Sulfuric acid	$(101, 0, 298.15)$	Absent from effective summary; first report coefficient MISSING
HEX	No cached entry	MISSING; named in substitution warning; CISOLID declaration
Apparent $NiSO_4$	No cached entry	MISSING in report
Five modeled ions	No cached entries	MISSING in generic report table

The archive retains the backup cache and effective-summary sections, report headers, warning text and source hashes in `outputs/aspen_revision_v24_20260928/property_audit_v24.json`. A paired calculation then supplies sulfuric-acid CPDIEC explicitly: either the water correlation $(A, B, C) = (78.54003, 31989.38, 298.15)$ or the cached ELECPURE acid correlation $(101, 0, 298.15)$. Each calculation starts from the same saved process inputs, clears and resolves the flowsheet, exports the effective property report, and verifies the saved parameter values after reopening. The report identifies the selected correlation as an INPUT value, establishing that the comparison changes the effective parameter record.

Across W700, P75-WASH and the high-acid MVR case, the paired choices give identical saved fresh-Ni, acetone-makeup, HEX, residual-Ni and product-acid values. The largest absolute block-duty differences are 4.2×10^{-10}, 4.2×10^{-9} and 4.2×10^{-8} MW, respectively. The corresponding ASC index differences are below 2×10^{-7} USD/h. All six calculations pass the numerical and

conservation audits. This isolated acid-dielectric choice therefore has no resolved effect on the reported comparisons at these states. The diagnostic uses fixed controller inputs; the original W700 feed retains its saved rounding relative to the nominal recipe. The queried stream mole fractions use apparent-component reporting, so they are not used to infer the molecular-acid fraction or explain the negligible response by speciation.

Table S4: Principal property and equipment assumptions used by the saved Aspen cases. Values are case inputs or literature-derived property entries.

Item	Input	Source / role
Global property framework	ENRTL–RK	Electrolyte process calculation
Original acetone–water NRTL	$a_{12} = -3.0768$, $a_{21} = 7.9385$; $b_{12} = 1203.7292$ K, $b_{21} = -2099.6736$ K; $\alpha = 0.3$	Aspen APV150 VLE–RK parameter set; 1 =acetone, 2 =water
Solid hydrate $\Delta_f G^\circ$	-2225.467 kJ/mol	OECD/NEA [3]
Solid hydrate $\Delta_f H^\circ$	-2683.817 kJ/mol	OECD/NEA [3]
Solid hydrate S°	334.450 J/(mol K)	OECD/NEA [3]
Solid hydrate C_p	$69.435 + 0.867T_K$ J/(mol K)	Kobylin [4]
Ni–acid–acetone SLE	700/820 mg kg^{-1} reference endpoints; additional complete cases from 600 to 1400 mg kg^{-1}	Conditional case endpoint; not an Aspen-fitted Ni mixed-solvent parameter
Fresh-feed preconcentration	50% of initial free water; additional 75% case	Case input; bulk-liquid aqueous phase screening reported in Section 2
Filter and cake	99.5% capture; 0.20 kg retained free liquid/kg crystal	Engineering inputs for solid and entrained-liquid inventories
Wash treatment	BASE/C: 0.40, 98%, 1.0%; W: 0.60, 99%, 1.5%; P75: 0.60, 99.05%, 2.0%	Wash ratio (kg/kg), displacement and dissolution; see Table S11
Primary acetone recovery	20 stages; 1.013 bar top; original-case reflux 0.02060–0.02684	Controller-set reflux; strengthened recovery targets 99.90/99.95%
Secondary acetone recovery	20 stages; 0.6 bar top; recovery and overhead-composition targets	Design specification
MVR compressor efficiencies	0.75 isentropic; 0.95 mechanical; 0.95 motor	Engineering scenario
Heat-exchanger coefficient	700 W m^{-2}K^{-1} base; 350–1000 sensitivity	Area calculation
Product desolventization	100 ppmw acetone; 0.05 wt% free water	Imposed outlet targets, separate from dryer thermal design

S1.3 Displacement and contact-time screen

A simple physical screen distinguishes accessible-pore displacement from dissolution time. For ideal piston washing with a stagnant-liquid fraction b, the largest displacement is $\eta \leq \min(N, 1-b)$, where N is incoming wash volume divided by initial retained-liquid volume, with constant pore volume and no volume change on mixing. The assigned wash-to-retained-liquid mass ratios are two for the aqueous wash and three for the stronger ASC wash. They equal N only under an equal-density approximation; actual wash and mother-liquor densities are needed for conversion. Once sufficient pore volumes are supplied, 98 and 99% displacement still require $b \leq 0.02$ and 0.01, respectively, plus sufficiently little bypass and dispersion. These are necessary hydraulic conditions for the assigned targets.

As a separate contact-time approximation, write $d(t) = d_\infty[1 - \exp(-k_d t)]$, where d_∞ is the crystal fraction that the contacting liquid could dissolve at its composition and temperature. Simultaneous operation requires the displacement target to be reached while

$$k_d t \leq -\ln(1 - d_{\max}/d_\infty), \qquad 0 < d_{\max} < d_\infty. \tag{S2}$$

Crystal surface area, particle size and mass transfer are absorbed into k_d. Wash breakthrough, cake hold-up and dissolved nickel measured over the same contact time determine whether the assigned displacement and loss can be attained together.

Hydrate water and conditional dissolution capacity. On a basis of one kilogram initial HEX contacted by w kilograms pure water, dissolving n moles transfers six waters per formula unit into the liquid. At an assumed saturation molality m_s,

$$n = m_s(w + 6M_w n), \qquad d_\infty = \frac{M_H m_s w}{1 - 6M_w m_s}, \tag{S3}$$

where M_w and M_H are the water and HEX molar masses in kg/mol. The expression applies while undissolved HEX remains and $6M_w m_s < 1$; available crystal mass limits $d_\infty \leq 1$. For the illustrative inputs $m_s = 2.6$ mol/kg water and $w = 0.40$ kg/kg, Eq. S3 gives 38.02% of the initial crystal mass; omitting hydrate water would give 27.34%. A 1% loss is 2.63% of this conditional capacity, and the original-parameter 7.37–8.02% parity range is 19.38–21.09%. The latter corresponds to 25.06–27.10% of the assumed saturation *molality*, because the water denominator also grows. This pure-water scenario excludes initial pore liquor, acid, phase conversion and contact kinetics. Its m_s is an illustrative assumption, not a measured HEX equilibrium for the process wash.

Composition and solid-phase evidence. Demirel et al. observed decreasing solution Ni with antisolvent addition in their tested compositions; the reported concentration uses initial aqueous volume [1]. Their samples were washed with pure organic solvent and dried at room temperature before XRD. O/A=2 samples gave tetragonal HEX, while an O/A=4 acetone experiment showed mixed hexahydrate forms; a lower-concentration sample aged for three weeks was tetragonal. These observations motivate composition, contact-time and phase measurements for the present washed cake. The 2.637 mol/kg $NiSO_4$, 0.133 mol/kg acid point transcribed by Sibarani et al. is assigned to the heptahydrate branch, whereas the selected 0.934/6.224 aqueous endpoint is assigned to α-HEX [2]. The low-acid point therefore supplies a separate phase reference rather than the pure-water HEX capacity in Eq. S3. The archive `wash_evidence_v24.json` records the source locations and the current washed-liquid Ni, acid and acetone inventories.

S1.4 Material-closure wash screens

The reference wash scans isolate the material consequences of cake hold-up, displacement and dissolution. Complete-flow resource comparisons are reported in the main text.

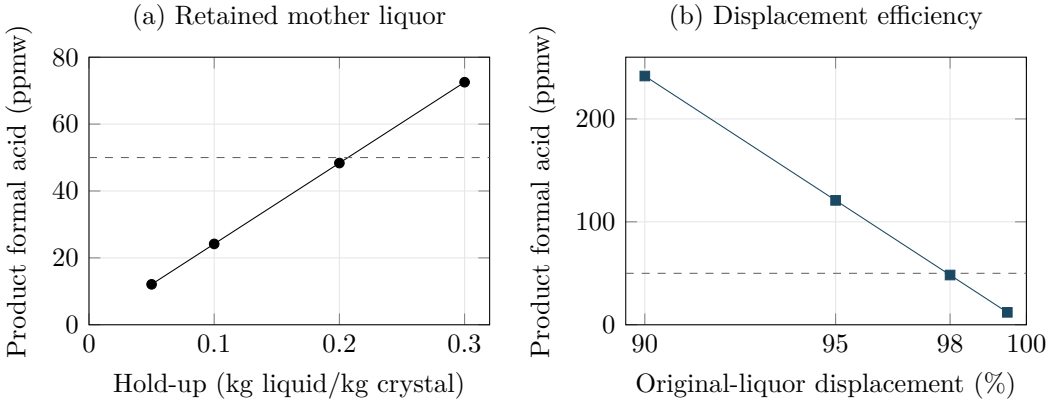


Figure S1: One-variable material-closure scans about BASE95. (a) Hold-up changes at 98% displacement; (b) displacement changes at 0.20 kg/kg hold-up. Other material assumptions are held fixed. Dashed lines show the 50 ppmw project target. Lines connect calculated scenarios; energy duties were not recalculated in these scans.

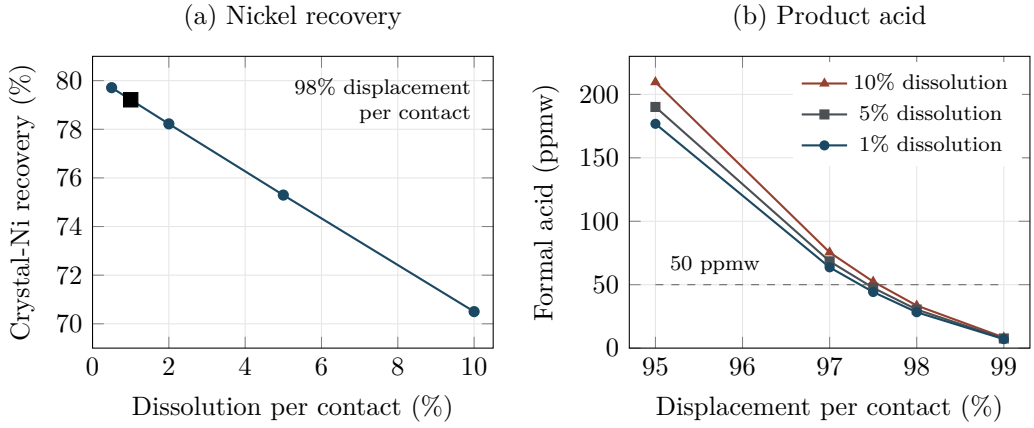


Figure S2: Two-contact aqueous-wash material scenarios at the selected high-acid endpoint and 1000 kg/h HEX output. (a) Recovery at 98% displacement; the black square denotes the wash specification used in the upstream-return EV/MVR cases. (b) Selected slices of the 25-point dissolution/displacement grid. Each contact uses 0.40 kg water/kg entering crystal with 0.20 kg/kg retained liquid. Lines connect the material-grid calculations; Fig. S11 separately reports utility recalculations at selected wash points.

S1.5 Acid-series material reconstruction

The complete-flow resource-index series uses upstream-return MVR at 0.05, 0.10 and 0.15 mol/L feed acid. Quadratic interpolation of the three ASC-minus-MVR differences gives parity at 0.1321 and 0.1352 mol/L for favorable and conservative ASC heat placement, respectively. The material-demand intersections in the following figure concern fresh Ni alone.

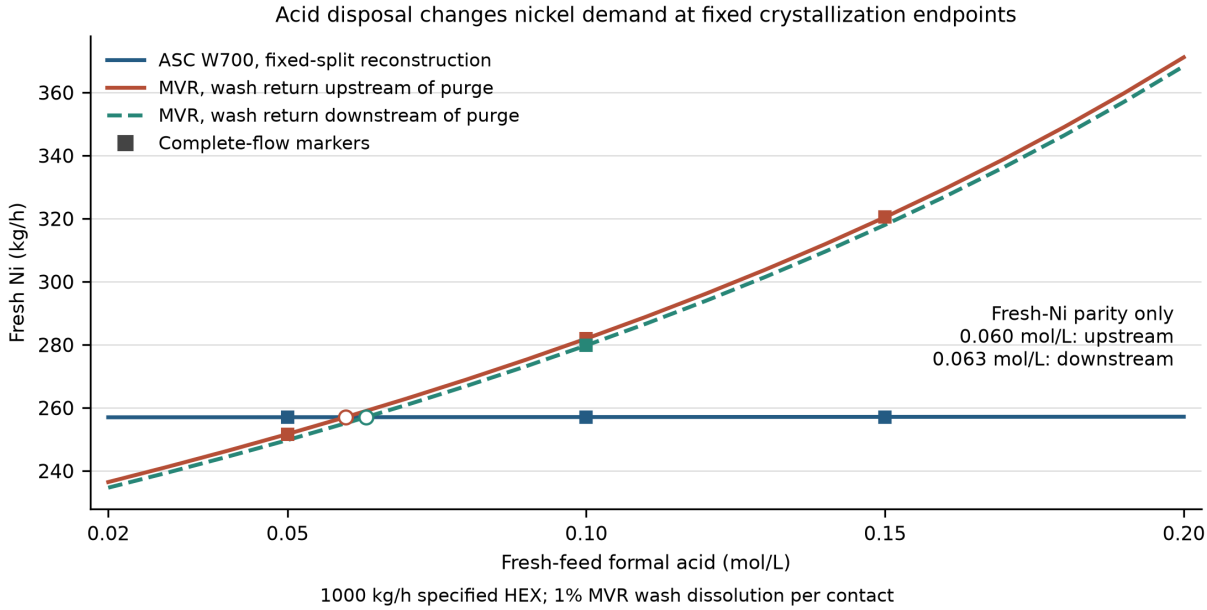


Figure S3: Conditional material-acid reconstruction at fixed crystallization endpoints and 1000 kg/h specified HEX. ASC retains observed W700 evaporator splits and overhead composition. MVR retains its aqueous endpoint and 1% dissolution per contact for each return arrangement. Filled squares are accepted complete-flow anchors; open circles mark fresh-Ni-demand parity, not resource-index parity.

S2 Native Aspen Plus process-flow diagram

Figures S4–S6 show the R95-W700 flowsheet in Aspen Plus V15. The display copy repositions labels and extends the drawn FINES outlet for legibility; calculation inputs, saved numerical results and physical stream connections are unchanged. The full view provides connectivity, while the enlarged regions identify equipment and streams. Stream values and equipment loads are reported in Tables S5–S6.

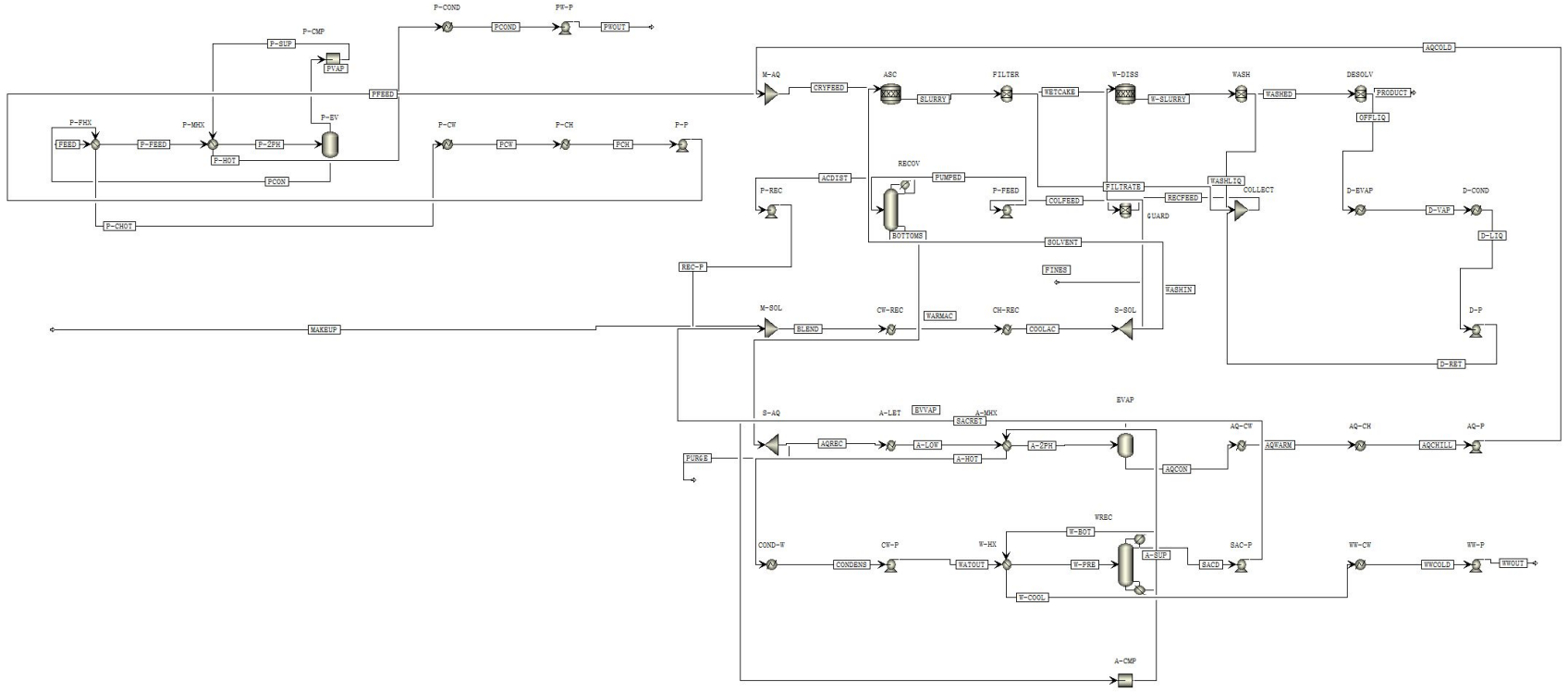


Figure S4: Native Main Flowsheet overview: 42 blocks and 58 material streams, captured at 78% canvas zoom. Fresh-feed concentration is at upper left, crystallization and primary solvent recovery at upper right, and aqueous recycle and secondary recovery below. Enlarged regions follow.

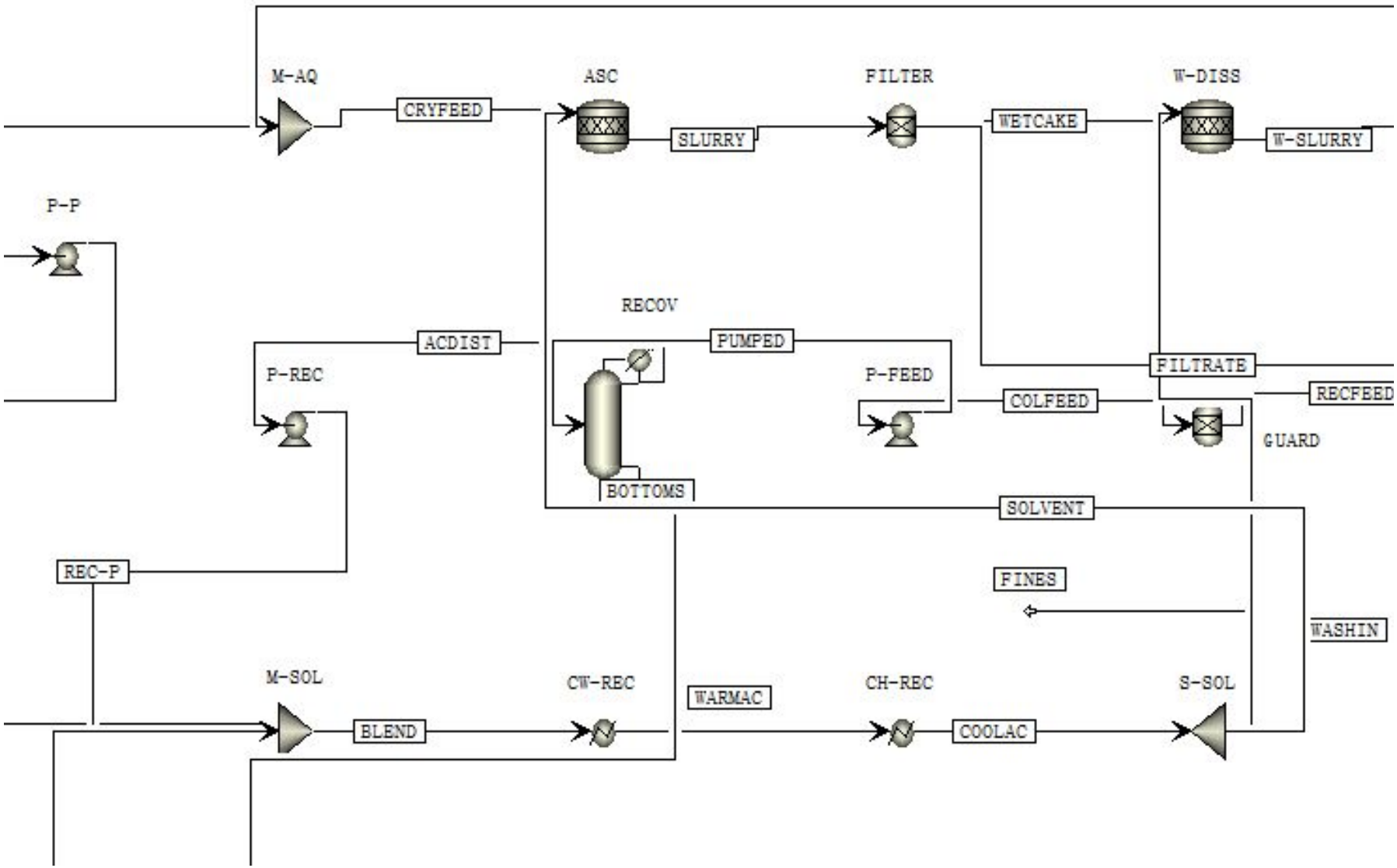


Figure S5: Native crystallization and solvent-recovery regions, shown in overlapping enlarged views. (a) Crystallization, filtration and primary acetone recovery. PFEED enters M-AQ from the left; AQCOLD returns from the lower aqueous circuit. RECOV overhead returns through solvent blending; BOTTOMS continues to S-AQ in Fig. S6. FINES is the external guard-solids outlet.

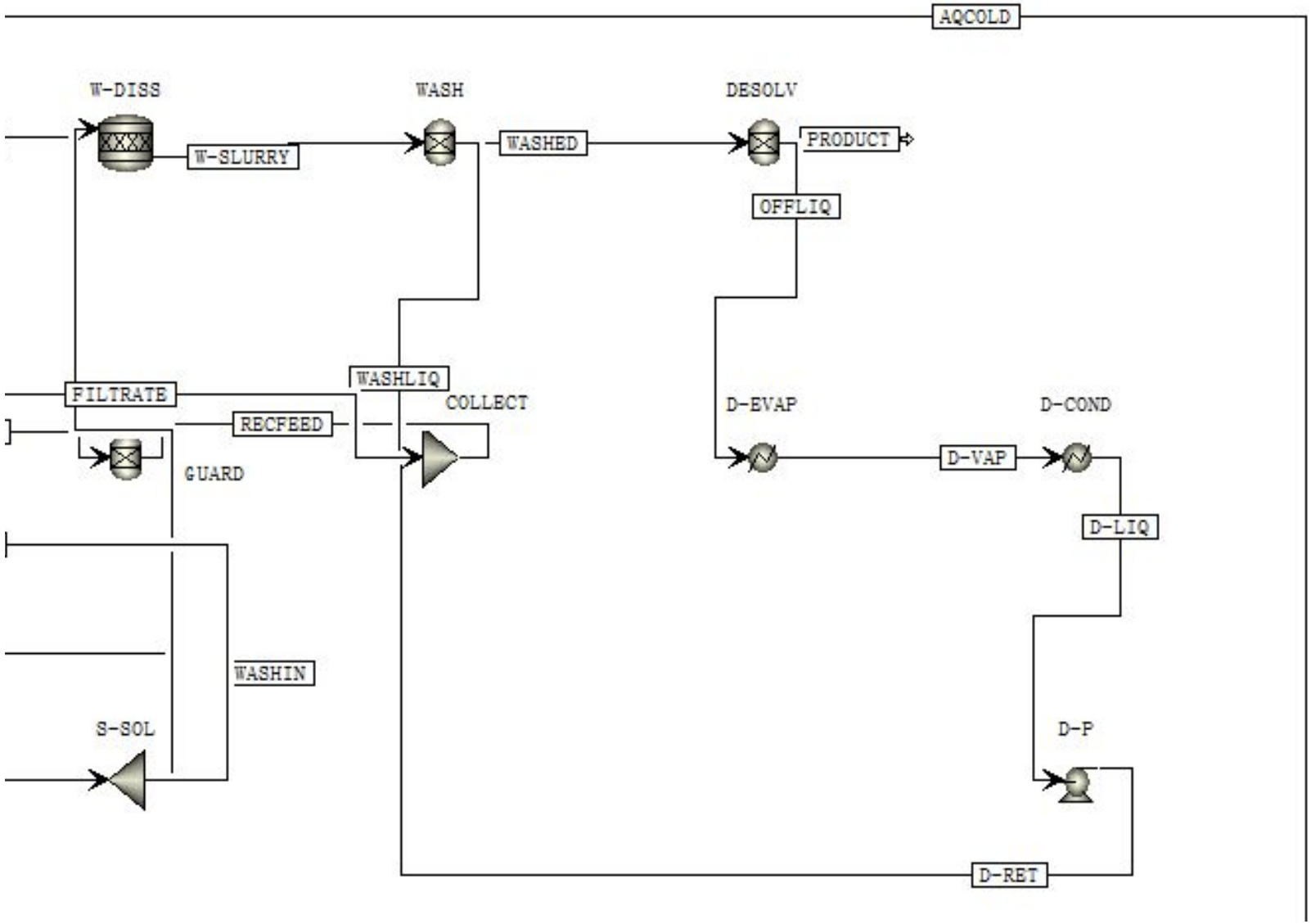


Figure S5: Native crystallization and solvent-recovery regions, continued. (b) Washing and desolventization. W-DISS and GUARD overlap the preceding view. WASHLIQ and FILTRATE join at COLLECT; the desolventization condensate returns through D-RET. The full connections are shown in Fig. S4.

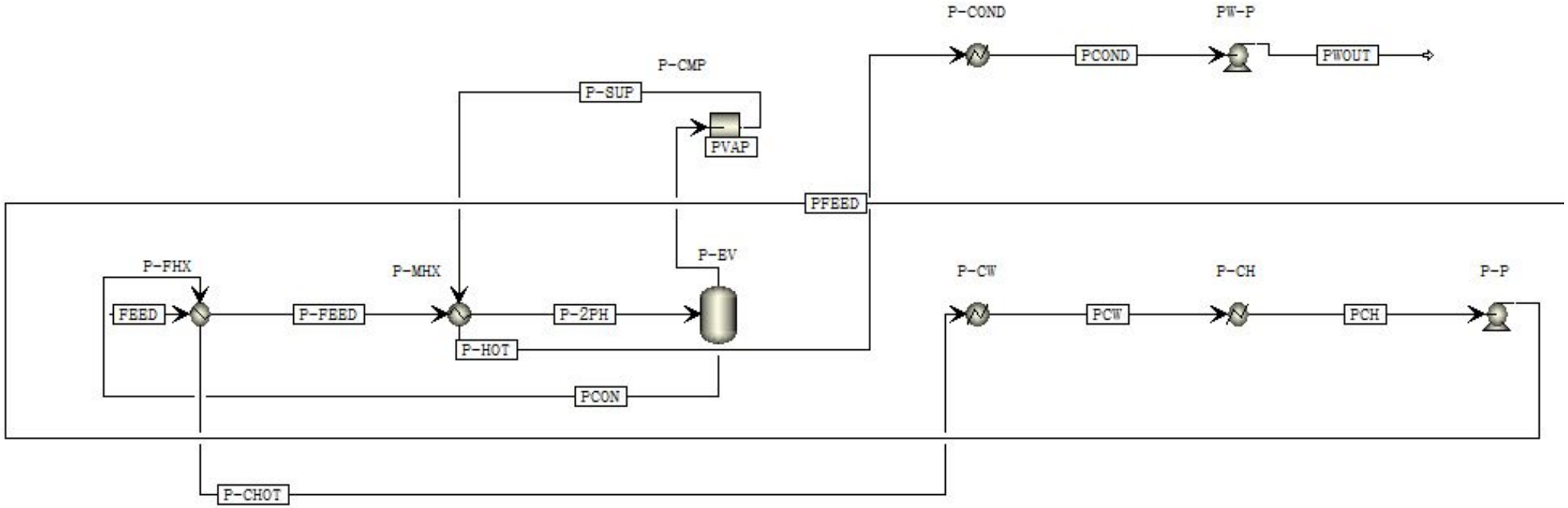


Figure S6: Native auxiliary regions. (a) Fresh-feed concentration and vapor recompression. P-EV removes fresh-feed water; PFEED continues to M-AQ. PWOUT is the final condensate outlet.

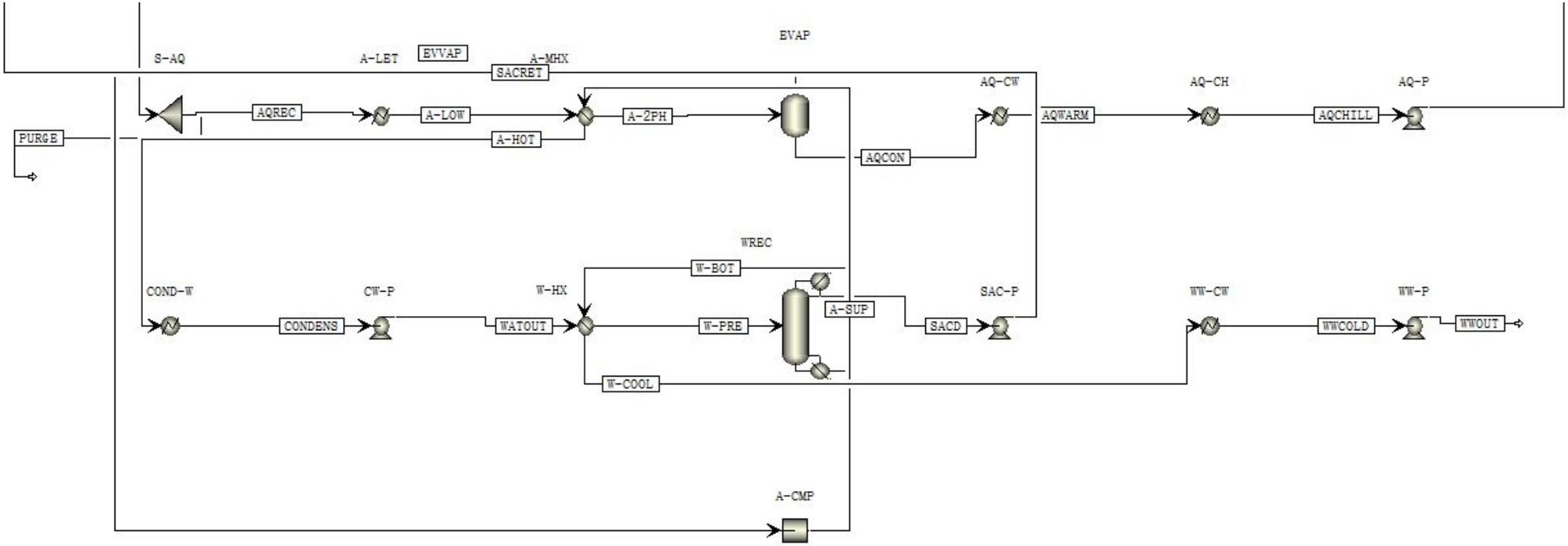


Figure S6: Native auxiliary regions, continued. (b) Aqueous recycle, purge and secondary acetone recovery. S-AQ receives RECOV bottoms. AQCOLD returns to M-AQ; WATOUT feeds WREC; SACRET returns recovered acetone to solvent blending. WWOUT is the final water outlet, and PURGE controls aqueous accumulation.

S3 Detailed reference stream and equipment inventory

The following records support the external resource allocation and connected heat-transfer analysis in the main text.

Table S5: Selected R95-W700 Aspen material streams ($kg\,h^{-1}$). Water denotes the free-water component; six stoichiometric waters in HEX are contained in its mass. Values are rounded from the saved stream snapshot; COLFEED is downstream of the fines guard.

Stream	Total	HEX	Apparent $NiSO_4$	Formal H_2SO_4	Free H_2O	Acetone
FEED	57698.516	0	677.730	554.811	56465.976	0
PFEED	29465.528	0	677.730	554.811	28232.988	0
CRYFEED	37179.858	0	760.845	1109.590	35305.557	3.866
SOLVENT	50591.782	0	0	0	3173.677	47418.105
SLURRY	87771.640	1020.330	160.112	1109.590	38059.636	47421.971
WETCAKE	1218.274	1015.228	0.375	2.597	89.081	110.994
PRODUCT	1003.475	1000.000	2.847	0.026	0.502	0.100
COLFEED	87371.792	0	166.231	1109.559	38103.430	47992.572
ACDIST	50849.927	0	0	0	3099.432	47750.495
BOTTOMS	36521.865	0	166.231	1109.559	35003.998	242.077
PURGE	18260.933	0	83.116	554.780	17501.999	121.039
PWOUT	28232.988	0	0	0	28232.988	0
WWOUT	10422.920	0	0	0	10417.061	5.859

Table S6: Selected R95-W700 recovery and heat-pump results. Column stage counts, compressor efficiencies and U are design inputs; duties, acetone recovery and suction volume are Aspen outputs; areas are integrated from the saved temperature–enthalpy profiles. A negative condenser duty is heat removed.

Service	Saved result	Design interpretation
RECOV, 20 equilibrium stages	Reboiler +14.558 MW; condenser −8.859 MW; 99.496% acetone recovery	47.750 t/h acetone in 93.905 wt% overhead
WREC, 20 equilibrium stages	Reboiler +0.265 MW; condenser −0.121 MW; 95.000% acetone recovery	111.314 $kg\,h^{-1}$ acetone in 90.000 wt% overhead
P-CMP / P-MHX	1.095 MW electric; 25.57 $m^3\,s^{-1}$ suction; 19.819 MW transfer; 1898.9 m^2	Fresh-feed water-removal MVR
A-CMP / A-MHX	0.529 MW electric; 9.50 $m^3\,s^{-1}$ suction; 6.270 MW transfer; 484.2 m^2	Recycle-water-removal MVR
P-FHX / W-HX	1.462 / 0.489 MW transfer; 98.4 / 60.8 m^2	Feed/condensate and secondary-recovery exchange

S4 Detailed resources, product material and thermal boundary

The following tables reconstruct the six reference indices and all design indices with their original endpoint heat allocation. For W700 and P75 at 700 mg/kg, the composition-specific condenser calculation in Table S34 supplies the refined absolute index used in the main text. O/C are alternative placements of unresolved phase-change heat; they are not statistical confidence limits. Tables S11–S14 give the design inputs, composition coordinates and resource terms; the archived ledgers retain full precision. Electricity is separated into compressor motor, modeled liquid pumps and estimated refrigeration; other auxiliaries are unquantified. Process cooling-water heat loads exclude the refrigeration condenser-side rejection and its cooling-water or fan services. Product inventories are reported for every original and additional complete case.

Table S7: Physical utility ledger (MW). Process CW includes the cooling-water duties charged in the index, excluding refrigeration condenser-side heat rejection.

Case	Heat	Process CW	Compressor	Pumps	Refrigeration
BASE95 C	16.610	14.204	1.635	0.004	0.477
BASE95 O	15.729	13.365	1.635	0.004	0.457
R95-C700 C	15.350	13.044	1.620	0.004	0.452
R95-C700 O	14.482	12.218	1.620	0.004	0.432
R95-W700 C	15.429	13.117	1.624	0.004	0.453
R95-W700 O	14.558	12.287	1.624	0.004	0.434
R95-W820 C	15.793	13.428	1.663	0.004	0.463
R95-W820 O	14.902	12.580	1.663	0.004	0.443
EV-WARM	34.994	28.438	0.000	0.001	2.190
MVR-WARM	0.000	1.210	2.313	0.001	0.082

Table S8: Selected-index contributions (USD/h). Electricity combines the three reported physical terms.

Case	Ni	Acetone	Heat	Electricity	CW	J
BASE95 C	2586.38	127.40	478.37	211.53	10.23	3413.91
BASE95 O	2586.38	127.40	453.00	209.52	9.62	3385.92
R95-C700 C	2563.82	113.75	442.08	207.60	9.39	3336.64
R95-C700 O	2563.82	113.75	417.07	205.60	8.80	3309.03
R95-W700 C	2570.43	114.50	444.35	208.14	9.44	3346.86
R95-W700 O	2570.43	114.50	419.27	206.16	8.85	3319.20
R95-W820 C	2631.03	117.18	454.83	212.98	9.67	3425.69
R95-W820 O	2631.03	117.18	429.19	211.00	9.06	3397.46
EV-WARM	2818.98	0.00	1007.84	219.10	20.48	4066.39
MVR-WARM	2818.98	0.00	0.00	239.53	0.87	3059.38

Table S9: Retained non-HEX inventories and total product mass. Each case specifies 1000 kg/h HEX, which is included in the total. Apparent $NiSO_4$ is nickel-bearing material of unresolved final hydrate state, not automatically a foreign impurity. All entries are kg/h.

Case	$NiSO_4$	Formal acid	Acetone	Free water	Total
BASE95	2.8215	0.0485	0.1003	0.5017	1003.4725
R95-C700	2.8215	0.0517	0.1003	0.5017	1003.4752
R95-W700	2.8466	0.0260	0.1003	0.5017	1003.4747
R95-W820	2.8472	0.0260	0.1003	0.5017	1003.4753
EV-WARM	2.8776	0.0284	0.0000	0.5017	1003.4077
MVR-WARM	2.8776	0.0284	0.0000	0.5017	1003.4077
MVR-D082	20.0309	0.0330	0.0000	0.5103	1020.5742
R95-W800	2.8471	0.0260	0.1003	0.5017	1003.4752
BASE95-ALT	2.8215	0.0485	0.1003	0.5017	1003.4720
R95-W700-ALT	2.8466	0.0259	0.1003	0.5017	1003.4746
R95-W820-ALT	2.8472	0.0259	0.1003	0.5017	1003.4753
R95-W700-P75	2.8466	0.0518	0.1004	0.5017	1003.5005
R95-W700-PURGE40	2.8466	0.0308	0.1003	0.5017	1003.4795
BASE600	2.8204	0.0485	0.1003	0.5017	1003.4710
BASE800	2.8225	0.0485	0.1003	0.5017	1003.4731
R95-W1000	2.8482	0.0259	0.1003	0.5017	1003.4762
MVR-D050	13.0851	0.0308	0.0000	0.5068	1013.6228
MVR-D100	23.5470	0.0344	0.0000	0.5120	1024.0934
ASC-ACID005	2.8466	0.0130	0.1003	0.5017	1003.4617
ASC-ACID015	2.8466	0.0389	0.1003	0.5017	1003.4876
ASC-P75-WASH	3.7652	0.0494	0.1004	0.5022	1004.4173
ASC-REC9990	2.8466	0.0260	0.1003	0.5017	1003.4747
ASC-REC9995	2.8466	0.0260	0.1003	0.5017	1003.4747
MVR-ACID005	2.8776	0.0284	0.0000	0.5017	1003.4077
MVR-ACID015	2.8776	0.0284	0.0000	0.5017	1003.4077
MVR-DOWN-D010	2.8776	0.0284	0.0000	0.5017	1003.4077
MVR-DOWN-D082	20.0309	0.0330	0.0000	0.5103	1020.5742
P75-C1100	3.7673	0.0494	0.1004	0.5022	1004.4193
P75-C1400	3.7688	0.0493	0.1004	0.5022	1004.4208
DOWN-D100	23.5470	0.0344	0.0000	0.5120	1024.0934

Table S10: Explicit volatile-treatment thermal loads (MW); signs follow heat into the process. These loads are included before heat allocation.

Case	DESOLV cut	D-EVAP	D-COND
BASE95	0.00011	0.03862	-0.04181
R95-C700	0.00011	0.03867	-0.04187
R95-W700	0.00011	0.03824	-0.04140
R95-W820	0.00011	0.03824	-0.04140
EV-WARM	0.00004	0.13493	-0.13950
MVR-WARM	0.00004	0.13493	-0.13950

S4.1 Specifications and resources of the design cases

The design cases share 1000 kg/h specified HEX. ASC-A005/A015 and MVR-A005/A015 denote the feed-acid cases. DOWN denotes aqueous wash return downstream of the purge split. O/C retain the common heat-allocation convention.

Table S11: Specified design inputs. Acid is mol/L feed; preconcentration, displacement, dissolution and primary recovery are percentages; c is mg Ni/kg final liquid. Dashes in the recovery column denote the reference recovery specification. MVR retains its aqueous crystallization specification.

Case	Acid	Preconc.	c	Displace.	Dissolve	Recovery
ASC-A005	0.05	50	700	99.00	1.5	–
ASC-A015	0.15	50	700	99.00	1.5	–
REC9990	0.10	50	700	99.00	1.5	99.90
REC9995	0.10	50	700	99.00	1.5	99.95
P75-WASH	0.10	75	700	99.05	2.0	–
MVR-A005	0.05	–	–	98.00	1.0	–
MVR-A015	0.15	–	–	98.00	1.0	–
DOWN-D010	0.10	–	–	98.00	1.0	–
DOWN-D082	0.10	–	–	98.00	8.2	–
P75-C1100	0.10	75	1100	99.05	2.0	–
P75-C1400	0.10	75	1400	99.05	2.0	–
DOWN-D100	0.10	–	–	98.00	10.0	–

Table S12: Composition coordinates for performance testing. Acetone is wt% of mother liquor excluding HEX; formal acid is mol/kg free water; Ni is mg/kg mother liquor. Product acid uses total product mass.

Case	Acetone	Acid	Ni	Product acid (ppmw)
ASC-A005	55.016	0.1478	700.0	12.95
ASC-A015	54.313	0.4484	700.0	38.79
REC9990	54.662	0.2974	700.0	25.89
REC9995	54.662	0.2974	700.0	25.90
P75-WASH	54.267	0.6048	700.0	49.23
MVR-A005	0.000	6.2240	31236.5	28.30
MVR-A015	0.000	6.2240	31236.5	28.30
DOWN-D010	0.000	6.2240	31236.5	28.30
DOWN-D082	0.000	6.2240	31236.5	32.35
P75-C1100	54.188	0.6043	1100.0	49.16
P75-C1400	54.130	0.6039	1400.0	49.11
DOWN-D100	0.000	6.2240	31236.5	33.55

Table S13: Physical resources and selected index for the design cases. Ni and acetone are kg/h; heat, total selected electricity and process cooling water are MW; J is USD/h.

Case	Ni	Acetone	Heat	Electricity	CW	J
ASC-A005 O	257.01	127.97	14.639	2.062	12.391	3322.0
ASC-A005 C	257.01	127.97	15.512	2.082	13.222	3349.7
ASC-A015 O	257.07	126.44	14.474	2.061	12.184	3316.3
ASC-A015 C	257.07	126.44	15.342	2.081	13.010	3343.8
REC9990 O	257.02	25.47	14.639	2.060	12.347	3229.7
REC9990 C	257.02	25.47	15.468	2.080	13.134	3256.1
REC9995 O	257.02	12.90	14.650	2.060	12.354	3218.6
REC9995 C	257.02	12.90	15.472	2.080	13.136	3244.9
P75-WASH O	241.76	60.32	6.850	2.007	6.386	2874.4
P75-WASH C	241.76	60.32	7.439	2.027	6.934	2893.8
MVR-A005	251.65	0.00	0.000	2.147	1.061	2731.9
MVR-A015	320.42	0.00	0.000	2.729	1.415	3478.1
DOWN-D010	279.76	0.00	0.000	2.400	1.216	3038.5
DOWN-D082	288.06	0.00	0.000	2.471	1.277	3128.6
P75-C1100 O	250.82	62.55	7.106	2.081	6.627	2982.0
P75-C1100 C	250.82	62.55	7.716	2.101	7.195	3002.0
P75-C1400 O	258.09	64.34	7.311	2.141	6.820	3068.4
P75-C1400 C	258.09	64.34	7.937	2.161	7.405	3088.9
DOWN-D100	289.78	0.00	0.000	2.486	1.290	3147.3

Table S14: Selected resource-index terms for the design cases (USD/h). Their sum reproduces J before rounding.

Case	Ni	Acetone	Heat	Electricity	CW
ASC-A005 O	2570.12	115.18	421.62	206.20	8.92
ASC-A005 C	2570.12	115.18	446.74	208.18	9.52
ASC-A015 O	2570.74	113.80	416.86	206.08	8.77
ASC-A015 C	2570.74	113.80	441.85	208.07	9.37
REC9990 O	2570.25	22.92	421.62	206.02	8.89
REC9990 C	2570.25	22.92	445.47	208.01	9.46
REC9995 O	2570.22	11.61	421.91	206.00	8.90
REC9995 C	2570.22	11.61	445.61	207.99	9.46
P75-WASH O	2417.58	54.29	197.29	200.67	4.60
P75-WASH C	2417.58	54.29	214.26	202.67	4.99
MVR-A005	2516.45	0.00	0.00	214.68	0.76
MVR-A015	3204.18	0.00	0.00	272.91	1.02
DOWN-D010	2797.61	0.00	0.00	239.99	0.88
DOWN-D082	2880.62	0.00	0.00	247.05	0.92
P75-C1100 O	2508.18	56.30	204.65	208.14	4.77
P75-C1100 C	2508.18	56.30	222.22	210.13	5.18
P75-C1400 O	2580.90	57.91	210.56	214.14	4.91
P75-C1400 C	2580.90	57.91	228.60	216.13	5.33
DOWN-D100	2897.83	0.00	0.00	248.57	0.93

S5 Pressure propagation and equipment allowance

Table S15: Saved MVR-WARM finisher states. Pressures are absolute; the total compressor ratio spans FINVAP to F-SUP. F-HX connects F-SUP to F-HOT, and F-COND provides final subcooling to FCOND.

Stream	Position	Pressure / mbar	Temperature / °C
FINVAP	Evaporator vapor / first-stage suction	19.6628	25.00
F-MID	First-stage discharge / second-stage suction	33.7705	81.53
F-SUP	Second-stage discharge / F-HX inlet	58.0000	147.90
F-HOT	F-HX liquid outlet at condensation endpoint	56.8871	35.00
FCOND	Subcooled liquid outlet	56.8871	34.00

The saved overall compressor ratio is $58.0000/19.6628 = 2.94973$. The F-HX inlet and outlet differ by 1.11289 mbar. Thus compression work uses the 58 mbar discharge state, whereas latent-heat matching uses the 35°C endpoint at 56.8871 mbar. These are distinct model nodes. The prescribed pressure change across F-HX has not been sized from exchanger or pipe hydraulics. The pressure-propagation calculation below retains the saved 58 mbar compressor discharge and the saved condensation endpoint.

The independent pressure calculation fixes the actual ternary liquid composition, 25°C crystallizer temperature and saved vapor mass rate, then perturbs the vapor-pressure boundary. Assumed overprediction magnitudes of 6.69 and 10.18% are borrowed from the pure-acid checks. They do not constitute a measured error interval for $NiSO_4$–H_2SO_4–H_2O at the selected endpoint. The evaporator pressure is $P_0/(1+\epsilon)$; suction loss is then subtracted. An isenthalpic suction loss gives the compressor inlet temperature. Two equal-pressure-ratio, uncooled compression stages retain efficiencies of 0.75/0.95/0.95. IF97 vapor-work increments are added to the original Aspen motor load, avoiding an artificial offset between the two property implementations. The change in condensation enthalpy and vapor-generation enthalpy is propagated into the heat cascade; liquid enthalpy and the aqueous SLE constraint remain fixed.

Table S16: Independent pressure-boundary scenarios, not new Aspen mixture-property calculations. The final row also imposes 3 mbar suction loss.

ϵ / %	Loss / mbar	P_s / mbar	Ratio	T_2 / °C	ΔE / MW	ΔJ / USD/h
0.00	0	19.66	2.950	147.9	0.000	-0.00
6.69	0	18.43	3.147	156.4	0.043	4.29
10.18	0	17.85	3.250	160.7	0.064	6.44
10.18	3	14.85	3.907	185.7	0.190	19.14

The zero-external-heat target persists in these fixed-composition scenarios at unchanged condensation pressures. The strongest case increases suction volume from about 594000 to 787000 m^3/h and selected electricity by 0.190 MW. This adds 19.14 USD/h to the index; it is much smaller than the 1%-wash MVR lead but matters near wash-induced parity. The calculation establishes the consequence of specified pressure errors, not their probability or actual magnitude. A validated high-acid ternary activity/enthalpy model and new complete mixture-property runs are still needed to close that thermodynamic uncertainty.

S5.1 Liquid-branch check for downstream wash return

Table S17 summarizes the evidence for the FEED-HX liquid branch at 8.2% dissolution per contact. The unrestricted temperature–heat calculation, the shared-stream enthalpy balance and the independent water-property calculation support sensible cooling. The hot inlet lies near saturation, so the local pressure drop remains a design variable.

Table S17: Phase and enthalpy checks for MVR-DOWN-D082. IF97 temperatures are calculated independently from the saved pressures. The unrestricted temperature–heat curves use inlet pressure.

Check	Result
Unrestricted hot-side curve	22 points, 85.000 to 37.753°C; vapor fraction zero at inlet pressure
Unrestricted cold-side curve	22 points, 25.000 to 65.000°C; vapor fraction zero at inlet pressure
Hot-stream composition	Water mass fraction > 0.999999
Hot inlet saturation margin	$T_{\mathrm{sat}} - T = -0.040$ K by IF97 at 0.57776 bar
Hot outlet saturation margin	46.413 K at 37.753°C and 0.560 bar
Independent sensible enthalpy	197.583 kJ/kg by IF97 versus 197.819 kJ/kg in Aspen; difference −0.119%
Rejected two-phase state	FEED-HX and B-COND had opposite energy residuals of approximately 6.1404×10^6 cal/s
Accepted shared-stream state	Both balances close at the audited tolerance; the saved cold rerun reproduces the liquid state and component flows

S6 External controller and verification record

The external controller is part of the model definition. Its full implementation, exact write paths, input units and per-iteration proposals are included in the archive. Its ASC solve order is:

1. Load a copied, identified BKP and record source hash, block/stream topology and the starting input/output state. Read the declared endpoint, solvent dose/purity, wash performance, fresh-water removal and purge fraction.

2. Solve the coupled apparent-component recycle balance using the last full-flow evaporator component splits, primary acetone recovery and actual overhead composition. The observed overhead acetone/water ratio closes makeup water when property parameters or composition change; the fixed original-overhead approximation is retained only in the explicitly algebraic endpoint grid. Preserve six waters per HEX. Scale fresh feed to 1000 kg/h final HEX after capture and dissolution. Solve solvent makeup, wash flow, conversion and component separation fractions.

3. Update feed and makeup rates, ASC conversion, filter/wash/desolventization splits, column distillate rates and wash split. Adjust primary and secondary reflux from successive recovery/purity observations using bounded secant updates; the secondary targets are 95% acetone recovery and 90 wt% overhead acetone. Adjust preconcentrator vapor fraction to the specified water removal and recycle-MVR transfer to a 0.05 MW evaporator trim.

4. Record each old value and requested value, then independently read back the changed leaves. Abort on a missing path or failed write/readback. Clear prior results before the first iteration; later iterations retain converged column estimates while changed inputs invalidate the affected results. Run the same owned Aspen document and export fresh stream, unit, status and report records.

5. Check global and block status, total/element balances, unit energy, six-water stoichiometry, phase and heat-exchanger approaches, then the performance targets. Repeat up to 25 controller iterations using the new full-flow observation. This is numerical matching to declared constraints, not optimization of their physical values.

6. Save with Save2, close the owned document, reopen the exact saved copy, check inputs and any alternative property parameters, clear results and recalculate. Compare all component flows to the final accepted run and preserve the reopened audit.

Key acceptance tolerances are 0.2 mg/kg for the endpoint, 0.05 kg/h for HEX output, 10^{-4} for dose factor, 2×10^{-5} for nominal solvent purity and wash/hold ratios, 0.001 kg/h for retained-wash component reconstruction, 0.0005 for secondary recovery and 0.0002 for secondary mass purity. The hot-trim tolerance is 0.001 MW. Relative element residuals must be below 10^{-6}; the ASC unit-energy tolerance is 10^{-4} MW. Interior exchanger approaches are checked at 10 K, with 0.01 K allowance for report rounding. Persistence compares component flows with $10^{-4} + 10^{-5}|m|$ kg/h tolerances.

For changed aqueous washing, the two-contact material solution sets fresh feed, aggregate dissolution, retention fractions, wash-condensate split, evaporation specifications and recycle guesses. The downstream-return D082 case fixes both FEED-HX side flashes to the liquid branch, supported by its temperature–heat checks and an independent IF97 comparison; this reproducible numerical setting is recorded in `phase_branch_check_v24.json`. It retains the prescribed physical production targets and does not establish strictly subcooled flow throughout installed equipment. The full Aspen run must independently recover the specified aqueous endpoint and 1000 kg/h HEX. Terminal phases, unit/plant conservation and compressor energy are checked before saving and reopening. Utility post-processing uses each newly calculated duty; historical extensive thermal curves are mass-scaled only for services whose composition, pressure and inlet temperature remain equal within recorded tolerances.

A failed numerical, phase or controller-target check stops the attempt and preserves its inputs, outputs and error. A COM timeout leaves status unknown and permits no additional command until its owner has completed. Failed numerical attempts are not admitted as solved cases. A successful reset/reopen test establishes numerical reproducibility at those inputs; experimental endpoint reachability, wash hydraulics and predictive accuracy outside the validated property region remain separate questions.

The product-acid screen is reported separately from numerical acceptance for the preconcentration and purge probes. A design probe that reproduces its specified input and material/energy solution is retained even when acid exceeds 50 ppmw, with a failed-screen label and no claim that it qualifies under the product requirement. The 75% preconcentration cases retain their respective wash and separation inputs and are recorded as separate configurations.

The P75 case with 99% displacement and 1.5% dissolution retained one cumulative simulation error: RECOV failed during recycle iteration 8, then recovered, and the recycle converged at iteration 59. For the separate accepted P75-WASH case, an early RECOV iteration limit also retained one simulation-error count in the cleared cold rerun. A later RECOV execution reached $\mathrm{Err/Tol} = 2.4244 \times 10^{-4}$ and the recycle converged at iteration 54. All final block, convergence and unit-operation statuses were zero, and the independent reopened RECOV energy residual was -2.14×10^{-7} MW. These recovered events remain in the history alongside the accepted final audits. The archived runtime property warnings accompany the controlled acid-dielectric comparison in Section S1.2. Control-panel events were not captured for BASE95-ALT, W800 and MVR-D082, so a zero-error history is not established for those runs. The machine-readable run-history review reports event coverage separately from final convergence and product screening.

For the alternative binary set, WREC's standard cold initialization diverged after an otherwise matched solution. The saved case was recovered using Aspen's NONIDEAL convergence preset, with all physical parameters and separation specifications retained. A separate standard-preset

BASE600 attempt failed during matching; its full record is retained and a new source copy uses NONIDEAL. The related BASE800 boundary case uses that same numerical preset. The preset and its independent readback are included in the revised backups. The matched run and subsequent saved/reopened/cleared run must pass the same audits. Post-processing also evaluates the resource index from the reopened output, so its numerical repeatability can be compared directly with the economic differences near parity. Across the twelve property, endpoint and washing cases, the largest absolute final-versus-reopened index difference is 0.000071 USD/h at recorded precision. The corresponding bound for the nine acid, routing and recovery cases and their fourteen resource rows is 0.000065 USD/h. The archive retains the component-flow comparisons and utility reconstructions behind these repeatability checks.

S6.1 Unresolved strengthened-recovery combinations

The following combinations target 99.95% primary acetone recovery and remain outside the accepted resource tables. The saved solver records locate the numerical failure; they do not establish physical infeasibility.

Table S18: Unresolved combined specifications and the recorded failure location. Complete console traces are preserved with each case.

Combination	Recorded failure
BASE95 + 99.95% recovery	RECOV RadFrac component balance failed; stage 20 dried, BOTTOMS flash lacked feed enthalpy, and the subsequent electrolyte property calculation stopped. Global convergence status was unavailable.
W820 + 99.95% recovery	The subsequent controller iterate failed in RECOV with stage-20 dry-up and missing BOTTOMS enthalpy, followed by a property stop and unavailable global status. Earlier converged iterates did not complete the target and persistence checks.
P75 + 99.95% recovery	RECOV and WREC exceeded RadFrac iteration limits; recycle block `$OLVER02` remained unconverged and the final mass/energy audit failed.

S7 Supporting endpoint and reference-route comparisons

S7.1 Paired endpoint and utility budgets

The paired material grid expresses the recovery allowance relative to the baseline endpoint (Fig. S7a). With no utility saving, the lower-dose route can tolerate approximately 49–62 mg kg^{-1} additional dissolved Ni because of its material-resource benefit at equal endpoints. At $S_U =$ 25 USD/h the allowance is 98–107 mg kg^{-1}; at 50 USD/h it is 146–151 mg kg^{-1}. These ranges cover baseline endpoints of 600–800 mg kg^{-1}. They state the utility saving required for a given loss in recovery. In the complete flowsheets, the actual utility-cost saving relative to BASE95 is 37.9–38.2 USD/h for W700 and 22.6–22.9 USD/h for W820, explaining why a constant utility credit would misrepresent the full-case slope.

The original-parameter crossing also depends on differential heating (Fig. S7b). An extra 0.25 or 0.50 MW charged to the candidate shifts the crossing to approximately 791 or 780 mg kg^{-1}; a 0.25 MW saving shifts it to 813 mg kg^{-1}. With a 0.50 MW saving the candidate remains cheaper at 820 mg kg^{-1}, so no crossing is assigned beyond that evaluated endpoint. These imposed heat budgets quantify how much thermal-model error would alter the decision. They are distinct from the alternative phase-change heat allocations in Table S29.

Checks away from the material-grid anchors. Two boundary pairs, $(c_B, c_C) = (600, 800)$ and $(800, 1000)$ mg/kg, were recomputed as complete Aspen flowsheets. The checks compare the material part of Eq. S4; utility budgets in the algebraic grid remain independently specified. The fixed-split reconstruction differs from the full material penalty by [-0.03, 0.02] USD/h across these two pairs. The agreement supports the material reconstruction at these two tested edges; the utility-saving budget remains an independent design input for the wider grid.

S7.2 Reference-route utilities and nickel valuation

EV-WARM avoids acetone circulation but requires 281.90 kg h^{-1} fresh nickel, compared with 257.04 kg h^{-1} for R95-W700. Its external hot-utility target is 34.99 MW, whereas the acetone case requires 14.56–15.43 MW. MVR changes the evaporation utility structure: compressed water vapor supplies heat to the process, reducing the ideal external-hot-utility target to zero

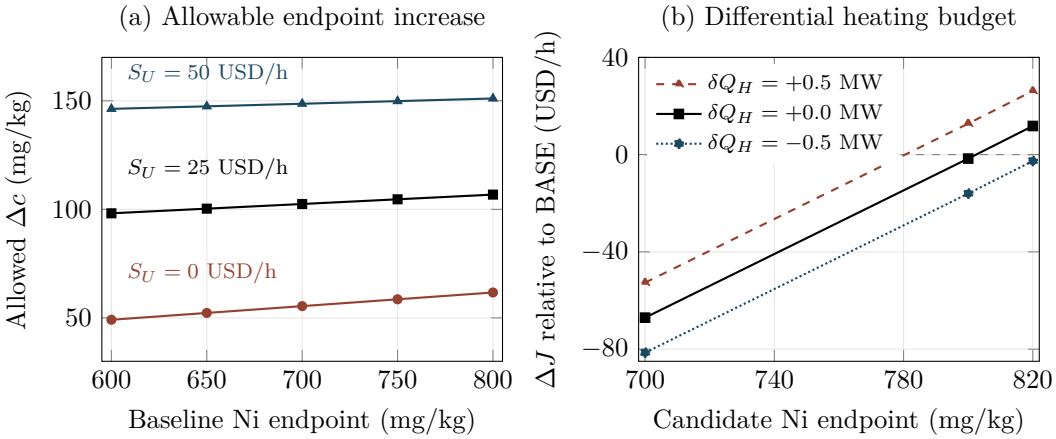


Figure S7: Decision requirements at the reference resource prices. (a) Paired material closures with locally fixed evaporator and recovery splits; S_U is an independent net utility-cost saving budget. (b) Piecewise-linear diagnostic through complete W700/W800/W820 cases, with imposed differential heating at the conservative allocation. The added heat costs are accounting scenarios. Negative ΔJ favors the candidate.

Table S19: Full-flow checks of two material-grid boundary pairs. Positive error means the fixed-split reconstruction underestimates the required utility saving.

c_B/c_C / mg/kg	Fixed splits	Full flowsheet	Difference
600/800	78.48	78.44	-0.03
800/1000	78.16	78.18	0.02

while selected electricity rises to 2.395 MW. This is heat substitution by electricity and internal recovery, not zero energy consumption.

At a 10 USD/kg fresh-Ni shadow value, MVR-WARM has a selected index 259.8–287.5 USD/h below R95-W700. The acetone case uses 24.86 kg h^{-1} less fresh nickel, so the index difference decreases as the assigned nickel-loss value increases. Holding the saved process states and other prices fixed, their selected-index parity occurs at 20.45–21.57 USD/kg fresh Ni (Fig. S8). At the nominal 10 USD/kg value, 259.8–287.5 USD/h of additional MVR-only cost would erase MVR's selected-index lead. This difference allowance is a useful equipment-economics target: vapor handling, heat-transfer area, vacuum service and route-specific investment can be evaluated against it rather than against a qualitative assertion of route superiority.

S8 Additional process specifications and property checks

The main comparison uses the feed recipe and output basis in the main-text Table 1. Nominal 95 vol% acetone is converted using pure-component reference densities of 0.784 and 0.997 kg/L. W700 uses an acetone-equivalent/free-water mass ratio of 1.34307381 at the aqueous crystallizer inlet; BASE95 uses 1.49230423. Feed preconcentration and downstream recycle determine the actual liquid composition.

S8.1 Closed-batch concentration normalization

The pure-salt conversion uses one initial litre, 1.00 mol Ni, 132.5 g total sulfate and an assumed 0.975 kg initial free water (sensitivity: 0.970–0.980 kg). The excess sulfate gives 0.3793 mol formal acid. Added acetone is $0.784(O/A)$ kg. All liquid Ni depletion is assigned to HEX, carrying six

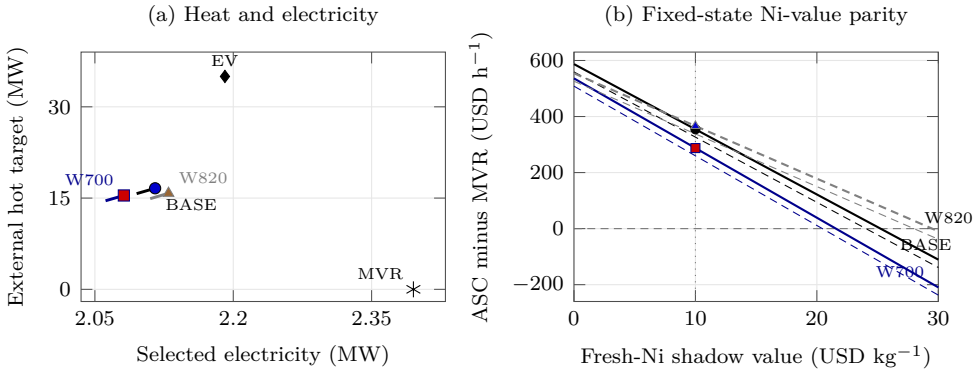


Figure S8: Route trade-offs at common HEX output. (a) Heat-cascade targets and selected electricity at the 10 K baseline. (b) Fixed-state ASC-minus-MVR resource index versus nickel shadow value; solid/dashed lines use conservative/optimistic ASC heat allocations. BASE, W700 and W820 parity ranges are 24.0–25.2, 20.5–21.6 and 28.0–29.5 USD/kg Ni.

Table S20: Aspen unit groups and the independently specified performance quantities in the antisolvent cases. Native block names identify the saved 42-block flowsheet.

Service	Representative native blocks	Model implementation and controlling quantity
Fresh-feed water removal	P-EV, P-CMP, P-MHX, P-COND	Remove 50% of feed water in the reference cases and 75% in a design probe; recompress vapor and transfer condensation heat.
Crystal production	M-AQ, M-SOL, ASC	Mix aqueous and antisolvent inlets at 25°C; impose final-liquid nickel endpoint and six-water solid stoichiometry.
Solid treatment	FILTER, WASH, W-DISS, DESOLV, GUARD	Specify capture, liquid retention, displacement and dissolution; meet assigned free-water and acetone outlet targets.
Primary solvent recovery	RECOV, CH-REC, CW-REC	Distill acetone-rich mother liquor; recycle overhead and split salt-bearing aqueous bottoms.
Aqueous water control	S-AQ, EVAP, A-CMP, A-MHX	Recycle part of aqueous liquor, purge the remainder and recompress water-removal vapor.
Secondary solvent recovery	WREC, W-HX, S-SOL	Recover acetone from the water-removal condensate before its aqueous outlet.

Table S21: Crystallization evidence and process endpoints. Literature liquid-Ni values are per initial aqueous volume; process endpoints are per final-liquid mass.

Record	Feed and addition at 25°C	Liquid Ni; crystal-Ni recovery	Model role
Pure-salt SLE [1]	1.00 mol/L Ni; pure acetone, O/A=2	0.02 mol/L (rounded); about 1.3 g/L in article text	Equilibrium series at a different feed composition
Post-IX B [1]	4.544 g/L Ni; pure acetone, O/A=2; 3 h	1.681 g/L; 62.10% collected	Measured operating endpoint
Post-IX E [1]	4.544 g/L Ni; staged pure/80%/pure acetone; 3 h	1.505 g/L; 54.06% collected	Measured staged-addition endpoint
Aspen W700/W820	0.100 mol/L formal acid; uniform 95 vol% acetone	0.700/0.820 g/kg; calculated crystal flow	Specified final-liquid endpoints

moles of water per mole Ni into solid. Thus, for printed residual Ni n in mol per initial litre,

$$m_{w,f} = m_{w,0} - 6(1-n)M_w, \quad m_{L,f} = m_{w,f} + m_{Ac} + nM_{NiSO_4} + n_{a,0}M_{H_2SO_4},$$

Table S22: Composition-specific experimental targets for the four principal ASC cases. Acetone is reported as wt% on both total-liquid and acetone-plus-free-water bases; HEX is excluded from liquid mass. Acid is mol/kg free water and Ni is mg/kg total liquid. All states are at 25°C.

Case	Ac / liquid	Ac / solvents	Acid molality	Ni endpoint
BASE95	57.032	57.833	0.29443	700
R95-C700	54.664	55.475	0.29724	700
R95-W700	54.664	55.476	0.29725	700
R95-W820	54.641	55.470	0.29716	820

Table S23: Experimental property checks. MAE denotes mean absolute error; y_{Ac} is vapor acetone mole fraction. Candidate values describe the alternative binary set used in the additional parameter-sensitivity cases; the original six cases retain the original set.

Quantity and subset	Points	Original MAE	Candidate MAE
Bubble temperature, all / K	72	0.433	0.516
y_{Ac}, all / mol/mol	72	0.00552	0.00900
Mixing enthalpy, all / J mol^{-1}	91	644.5	100.0
Mixing enthalpy, withheld / J mol^{-1}	39	719.2	145.5

Table S24: Aqueous endpoint selection: material screening with two wash contacts. Only the last row was subsequently solved as the EV and MVR complete flowsheets. Molalities use free water.

$m_{H_2SO_4}$ mol/kg	m_{NiSO_4} mol/kg	Fresh feed t/h	Crystal-Ni recovery %	Product acid ppmw
3.154	1.409	120.61	41.56	16.5
5.215	1.020	68.26	73.44	24.9
6.224	0.934	63.28	79.21	28.3

Table S25: Six original saved flowsheet cases. Additional cases change endpoint, binary parameters, preconcentration, purge or MVR dissolution as specified in the results. Dose factor multiplies the stated BASE95 pure-acetone/free-water ratio. ASC uses a specified residual Ni; the aqueous value is calculated from its selected molality pair. Wash ratios refer to crystal entering each contact.

Case	Dose factor –	Residual Ni $mg\,kg^{-1}$	Wash ratio kg/kg crystal	Displacement %	Wash dissolution %
BASE95	1.00	700	0.40	98	1.0
R95-C700	0.90	700	0.40	98	1.0
R95-W700	0.90	700	0.60	99	1.5
R95-W820	0.90	820	0.60	99	1.5
EV-WARM	–	31237	2×0.40	98 per contact	1.0 per contact
MVR-WARM	–	31237	2×0.40	98 per contact	1.0 per contact

with molar masses in kg/mol. The horizontal coordinate is $100m_{Ac}/(m_{Ac}+m_{w,f})$ and the vertical coordinate is $10^6 nM_{Ni}/m_{L,f}$.

For B/E, the vertical coordinate uses the reported 4.544 g initial Ni, 1.681/1.505 g residual Ni, an assumed initial solution density of 1.020 kg/L, and pure-HEX liquid depletion. Added acetone/water volumes are 2.00/0 and 1.90/0.10 L. Their free-water inventory is unreported, so the horizontal coordinate uses a separate stated convention of 1.000 kg initial free water per litre. No process-feed acid inventory is assigned to the experiments. Sampling, evaporative losses and impurity-bearing solids are unallocated.

Table S26: Literature endpoints and process targets on common axes. Acetone is wt% of acetone plus free water; Ni is mg/kg total final liquid. Literature values are conditional mass conversions of Demirel et al. [1], not new measurements. B/E use the stated initial-water convention for the acetone coordinate.

Point	Acetone (wt%)	Ni (mg/kg)	Acid (mol/kg water)
Pure salt O/A=0.5	30.31	13608	0.421
Pure salt O/A=1	47.15	3760	0.432
Pure salt O/A=2	64.34	474	0.436
Post-IX B	61.18	653	Unreported
Post-IX E	57.65	580	Unreported
W700 target	55.48	700	0.297
P75 target	55.79	700	0.605

Log-linear interpolation between the pure-salt O/A=1 and 2 points gives 1.33 g Ni/kg at P75's 55.79 wt% solvent-basis acetone. A separate illustration takes the B concentration as an anchor and assumes $\ln c = \ln c_B + s(x_B - x)$, with x in acetone weight-percentage points and $s = 0.08$–0.12 per percentage point. It gives 1.00–1.25 g/kg at the P75 coordinate. These two calculations are displayed as magnitude comparisons across different solution matrices. The low printed pure-salt concentration (0.02 mol/L) has about 25% relative half-digit rounding uncertainty; pure-salt acid molality is about 0.43 mol/kg, post-IX acid is unknown, and each experiment's addition and equilibration history remains part of its interpretation.

S9 Reference solvent reduction and washing comparisons

S9.1 Complete-flow reference and dose cases

The inventory and duty tables identify two different optimization levers. RECOV dominates gross heating, so its reflux, pressure and recovered-solvent composition govern the thermal

Table S27: Complete-flow results at 1000 kg h^{-1} modeled hydrate production. Hot utility is a heat-cascade target; Aspen gross heating is a different measure. For ASC, intervals are the optimistic–conservative thermal-grade allocations.

Case	Fresh Ni kg/h	Crystal Ni recovery, %	Acetone makeup, kg/h	Aspen gross heating, MW	External hot utility, MW	Selected electric incl. refrig., MW	Product acid ppmw
BASE95	258.64	86.34	141.56	16.65	15.73–16.61	2.095–2.115	48.35
R95-C700	256.38	87.10	126.39	15.39	14.48–15.35	2.056–2.076	51.50
R95-W700	257.04	86.87	127.22	15.47	14.56–15.43	2.062–2.081	25.88
R95-W820	263.10	84.87	130.20	15.83	14.90–15.79	2.110–2.130	25.87
EV-WARM	281.90	79.21	0	43.32	34.99	2.191	28.30
MVR-WARM	281.90	79.21	0	40.45	0 (ideal cascade)	2.395	28.30

burden. Yet improving WREC from its specified 95% recovery to complete recovery could save at most the 5.859 kg h^{-1} acetone now leaving in WWOUT at this fixed process state, whereas the aqueous purge carries 121.039 kg h^{-1}. Lower purge may retain more nickel and acetone but also increases acid, impurity and water circulation. A plant-wide solvent study must therefore vary crystallization endpoint, solvent quality and purge together; a recovery-column efficiency alone is an incomplete objective.

The closed-loop BASE95 case requires 58.06 t h^{-1} of fresh liquor and 141.56 kg h^{-1} of makeup acetone to produce 1000 kg h^{-1} of modeled hydrate. Crystal-Ni recovery, defined as nickel in the modeled hydrate divided by fresh-feed nickel, is 86.34%. Primary and secondary acetone recovery, aqueous purge, recycle-water removal and product desolventization are all active in this solution. The plant-wide makeup requirement is determined by all solvent exits, not by the recovery percentage of the primary column alone.

With the mother-liquor nickel endpoint prescribed at 700 mg kg^{-1}, reducing added acetone by 10% in R95-C700 lowers makeup acetone to 126.39 kg h^{-1}, fresh nickel from 258.64 to 256.38 kg h^{-1}, and Aspen gross heating from 16.65 to 15.39 MW. The formal-acid concentration of its final product, however, is 51.50 ppmw, 1.50 ppmw above the project screen. BASE95 itself lies only 1.65 ppmw below that screen. The result makes product entrainment a consequential design variable, particularly because wash displacement is an assigned equipment-performance value.

The linked washing change in R95-W700 lowers formal product acid to 25.88 ppmw. Makeup acetone is 127.22 kg h^{-1}, 10.13% below BASE95, while gross Aspen heating is 15.47 MW, 7.10% below BASE95. Fresh nickel demand is 257.04 kg h^{-1} and crystal-Ni recovery is 86.87%. The strengthened wash carries 1.5% crystal dissolution instead of 1.0%, and the full circuit includes that returned nickel and the additional liquid. The resource saving is therefore retained at the prescribed 700 mg/kg endpoint after charging the stronger wash with additional liquid and dissolution. Table S27 compares the cases; Fig. S1 isolates washing effects.

Figure S9 explains the endpoint reversal. Moving from W700 to W820 adds 60.6 USD/h in fresh-nickel charge, 2.7 USD/h in acetone and 15.5 USD/h in combined utilities. Nickel loss is the largest contribution, but the utility change also erodes the lower-dose benefit. The development requirement is therefore to preserve recovery at the actual reduced-dose composition while retaining the associated utility saving.

At a 700 mg kg^{-1} mother-liquor endpoint, the reduced-dose W700 case lowers the selected resource index by 66.7–67.0 USD/h relative to BASE95. Raising the endpoint to 820 mg kg^{-1} adds 6.06 kg h^{-1} fresh nickel, lowers crystal-Ni recovery from 86.87 to 84.87%, and puts the index 11.5–11.8 USD/h above BASE95. The full-flow W800 point lies close to parity, with $\Delta J = [-1.77, -1.62]$ USD/h. The resource contributions in Fig. S9 show how the increasing nickel requirement offsets solvent and utility savings.

Table S28 compares each reduced-dose case with the BASE95 reference calculated using the same parameter set. The alternative acetone–water NRTL set retains a 61.1–61.3 USD/h W700 advantage and places W820 16.3–16.6 USD/h above its reference. Piecewise interpolation of the original 700/800/820 points gives parity near 803 mg kg^{-1}; the alternative 700/820 points

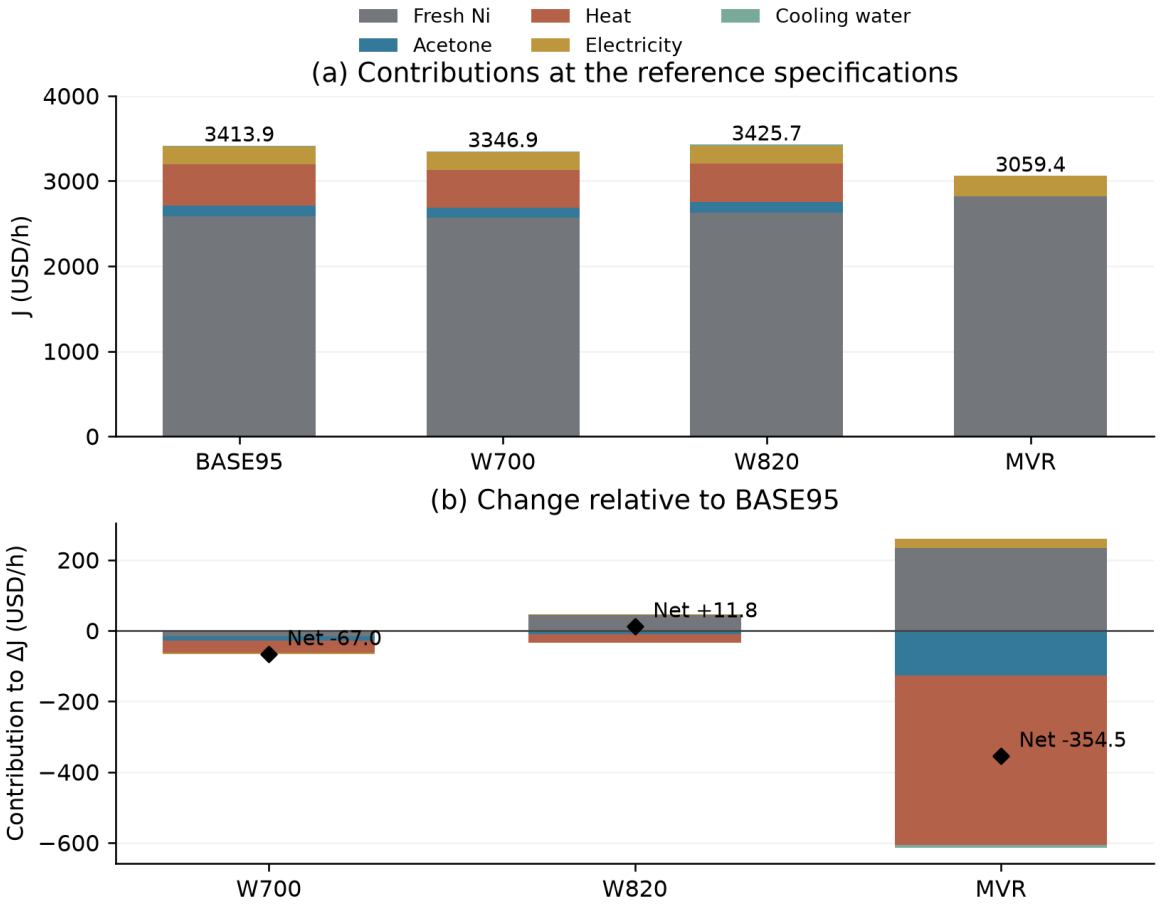


Figure S9: Resource-index contributions at the common prices. ASC uses the conservative heat allocation and MVR the profiled 10 K baseline. (a) The same five terms reconstruct each total. (b) Signed contributions relative to BASE95; black diamonds mark their sums. These are the reference 1%-per-contact MVR wash and the stated ASC wash cases, not route-wide economic rankings.

give about 795 mg kg^{-1} (Fig. S10a). Both sets therefore preserve the benefit at 700 mg kg^{-1} and remove it at 820 mg kg^{-1} under matched heat allocation. Relative to the 700 mg kg^{-1} reference, they allow approximately 95–103 mg/kg endpoint deterioration under their respective assumptions.

The operative design test is whether crystallization at the reduced-dose recycle composition can meet this endpoint requirement. Independent heat placements for candidate and reference give a wider comparison: the W820 difference spans both signs (Table S29 and Fig. S10b). Parameter dependence, heat-allocation dependence and experimental endpoint reachability are therefore distinct parts of the design decision.

Table S28: Full-flow parameter propagation at matched product and operating specifications. Each candidate is compared with BASE95 calculated using the same binary parameter set. O/C denote optimistic/conservative heat allocation; negative values favor lower dose.

Parameter set / endpoint	ΔJ_O / USD/h	ΔJ_C / USD/h
Original, 700	-66.72	-67.05
Original, 800	-1.77	-1.62
Original, 820	11.53	11.78
Alternative, 700	-61.06	-61.33
Alternative, 820	16.28	16.58

Interpolated parity (O/C), mg/kg liquid: original 802.67/802.42; alternative 794.74/794.46. These ranges compare heat allocations within each parameter set.

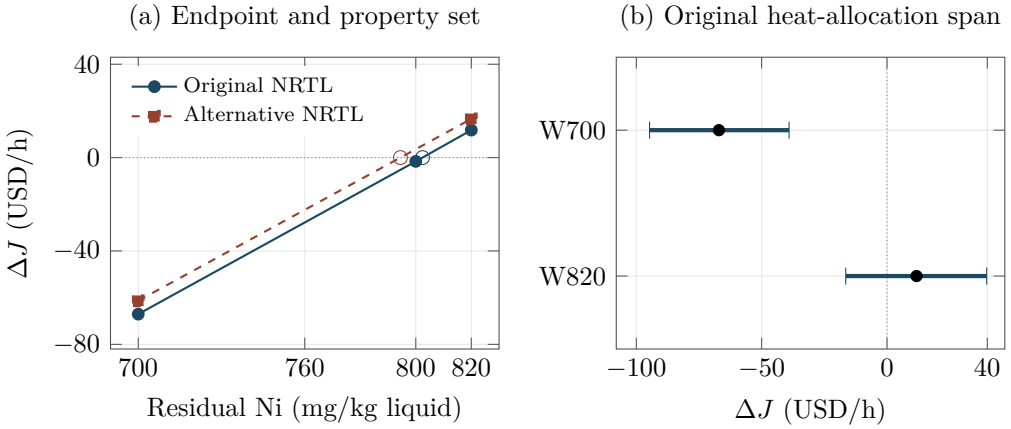


Figure S10: Residual-nickel requirements for the 10% lower-dose ASC configuration at the common reference prices. (a) Filled markers are complete-flow calculations, paired with BASE95 using the same NRTL set and conservative heat allocation. Segments interpolate within the sampled interval; open circles mark index parity. (b) Original-parameter spans allow independent heat placements for W700/W820 and BASE95; black points use matched conservative placement. Negative ΔJ favors lower dose. The output basis is 1000 kg/h specified HEX with zero credit for retained non-HEX salt.

Section S7.1 reports the paired-endpoint utility budgets and fixed-state heat perturbations that support these requirements.

S9.2 Material-grid calculation

To separate the baseline crystallization assumption from the penalty of lowering solvent dose, an additional material grid pairs five BASE95 endpoints, c_B = 600, 650, 700, 750 and 800 mg kg^{-1}, with reduced-dose endpoints $c_C = c_B + \Delta c$, where Δc = 0–200 mg kg^{-1} in 10 mg kg^{-1} steps. The

Table S29: Selected operating/resource index change relative to BASE95 at the common illustrative prices. Negative values favor the candidate on the selected terms. The formal-acid screen is applied independently of cost.

Case	Matched heat allocation	Independent heat bounds	Acid screen
	Index difference, USD/h		($\leq$ 50 ppmw)
R95-C700	−77.3 to −76.9	−104.9 to −49.3	Fail
R95-W700	−67.0 to −66.7	−94.7 to −39.1	Pass
R95-W820	+11.5 to +11.8	−16.5 to +39.8	Pass

105 pairs retain their respective BASE95 and strengthened-wash settings. Each route uses its own evaporator component splits and primary acetone recovery from the corresponding 700 mg kg^{-1} Aspen state. At the two 700 mg kg^{-1} anchors, the reconstructed fresh-Ni and acetone-makeup rates differ from the full-flow records by less than 0.003 kg h^{-1}.

For each pair, the minimum net utility-cost saving required to offset its material penalty is

$$S_{\mathrm{req}} = p_N(m_{N,C} - m_{N,B}) + p_A(m_{A,C} - m_{A,B}), \qquad \Delta J = S_{\mathrm{req}} - S_U, \tag{S4}$$

where S_U is the saving in the combined hot-utility, electricity and cooling-water charge. Budgets of 0, 25 and 50 USD/h define design requirements independently of a utility prediction for each material-grid point. The original W700/W820 cases provide utility-saving reference values; additional complete-flow checks test the crossing and selected grid edges. A separate fixed-state stress calculation adds a differential hot-utility load $\delta Q_H = -0.50$ to $+0.50$ MW to the full-case index difference; at the reference heat price this adds $28.8\delta Q_H$ USD/h. Cost crossings are interpolated only within the evaluated 700–820 mg kg^{-1} interval.

S9.3 Washing and product accounting

The material scans use a closed recycle algebra with locally fixed evaporator splits; the named R95 cases separately test representative points in the full Aspen process. Figure S1 shows actual responses rather than only parameter ranges. Increasing retained mother liquor from 0.05 to 0.30 kg/kg raises formal product acid from 12.1 to 72.5 ppmw. At 0.20 kg/kg hold-up, displacement of 90–99.5% gives 241.7–12.1 ppmw. Linear interpolation in these scans places the 50 ppmw requirement near 97.93% displacement, or approximately 0.207 kg/kg maximum hold-up at 98% displacement. These are conditional washer-performance requirements.

The BASE and C700 values of 48.35 and 51.50 ppmw straddle an analyst-selected acid target by small margins. Reporting acid continuously makes the implication clearer: W700 reaches 25.88 ppmw under its linked 0.60 kg/kg wash, 99% displacement and 1.5% dissolution settings. Increasing wash flow alone has little effect in the independent-parameter scan because displacement is specified separately. The linked case therefore represents an intensified wash service whose performance must be achieved by the equipment.

W700 wash liquid is the recycled-solvent blend, and the washed cake retains 0.20 kg liquid/kg final HEX. It contains 14.99 kg/h free water, 182.14 kg/h acetone and 2.847 kg/h apparent $NiSO_4$ before desolventization. DESOLV cuts the assigned volatile inventory into OFFLIQ; D-EVAP vaporizes that stream and D-COND condenses it. Both thermal duties enter the reported utility boundary (Section S4). Capture of 98–99.9% changes fresh-Ni demand from 262.6 to 257.6 kg/h in the BASE material scan, while 0.1–2% wash dissolution changes it from 256.9 to 260.5 kg/h. Both affect resource use even where the acid target is met.

Simultaneous wash performance. The relevant equipment test measures mother-liquor breakthrough and dissolved nickel over the same contact time. The 98–99% displacement targets require low bypass and stagnant-liquid retention; the assigned 1–1.5% dissolution losses require

sufficiently short contact or low dissolution capacity. Section S1.3 gives a simple pore-volume and contact-time screen for designing that test. It identifies measurements needed to assess each wash pair (η, d).

The W700 washed cake illustrates the composition to be tested: its non-HEX material contains 91.07 wt% acetone and 5.398 g Ni/kg, compared with 54.66 wt% acetone and 0.700 g Ni/kg in the crystallizer mother liquor. Formal-acid molality also changes from 0.29725 to 0.01766 mol/kg free water. Thus the prescribed dissolution and liquid displacement create a different acid–solvent composition, whose dissolved-Ni level and solid phase must be measured together. Literature experiments establish antisolvent precipitation and composition-dependent hydrate forms; Section S1.3 connects that evidence to the required wash measurements.

Aqueous wash. The two-contact aqueous wash has a different resource response from the solvent wash (Fig. S2). At 98% displacement per contact, increasing dissolution from 0.5 to 10% raises fresh-Ni demand from 280.1 to 316.7 $\mathrm{kg\,h^{-1}}$ and lowers crystal-Ni recovery from 79.7 to 70.5%. Between the reference 1% and the 5% scenario, fresh-Ni demand increases by 14.67 $\mathrm{kg\,h^{-1}}$, equivalent to 146.7 USD/h at the stated shadow value. The material grid isolates nickel demand. Complete recalculations at 5, 8.2 and 10% dissolution now include the changed feed flow and utility loads (Fig. S11).

Acid removal remains strong over the same dissolution range. At 98% displacement, modeled product formal acid is 28.1–33.5 ppmw; with 1% dissolution, lowering displacement from 98 to 95% raises acid from 28.3 to 176.8 ppmw. Solving for the 50 ppmw screen requires 97.33–97.56% displacement per contact across the 0.5–10% dissolution range. Two consecutive contacts reduce original-liquor carryover approximately with $(1 - \eta)^2$, while dissolution raises the upstream crystal-production requirement. Consequently, passing the acid screen alone does not constrain the recovery penalty. A water-wash design needs both measured displacement and dissolution at the chosen contact conditions.

Coupling wash loss to the reference route comparison. Complete-flow MVR calculations at 1, 5, 8.2 and 10% dissolution per contact give $J = 3059.4$, 3218.0, 3354.6 and 3436.1 USD/h. Original-parameter ASC W700 spans 3319.2–3346.9 USD/h; the alternative acetone–water set lowers this absolute range to 3276.1–3303.7 USD/h. These comparisons retain the reference MVR wash-liquor return upstream of the 50% purge split.

The 8.2% case retains the high-acid endpoint, 98% displacement and 1000 $\mathrm{kg\,h^{-1}}$ HEX output. It requires 309.20 $\mathrm{kg\,h^{-1}}$ fresh nickel and 2.617 MW selected electricity, with zero external heat at the prescribed 10 K match. Its index exceeds original-parameter W700 by 7.8–35.4 USD/h while product formal acid is 32.35 ppmw. Acid removal therefore remains effective as the nickel loss consumes the resource margin. Each complete-flow point uses its own Aspen duties; thermal profiles are rescaled only for services with verified matching composition, pressure and temperature.

At the 700 mg/kg endpoint, interpolated wash parity is 7.37–8.02% per contact with the original ASC parameters and 6.36–7.01% with the alternative set. The alternative binary parameters therefore reduce the wash loss required for ASC to recover its index advantage on the declared product basis.

Product-basis sensitivity. The original upstream-return 8.2% MVR case retains 20.031 kg/h apparent non-HEX $NiSO_4$, compared with 2.847 kg/h in original-parameter W700. This adds 6.517 kg/h product nickel outside specified HEX. At fixed HEX output, an assumed credit of 1.19–5.44 USD/kg for this additional Ni offsets the index penalty (Fig. S12). Product definition therefore belongs alongside crystallization and washing as a condition of the comparison. A separate denominator diagnostic scales all extensive MVR terms to W700's 224.380 kg/h total

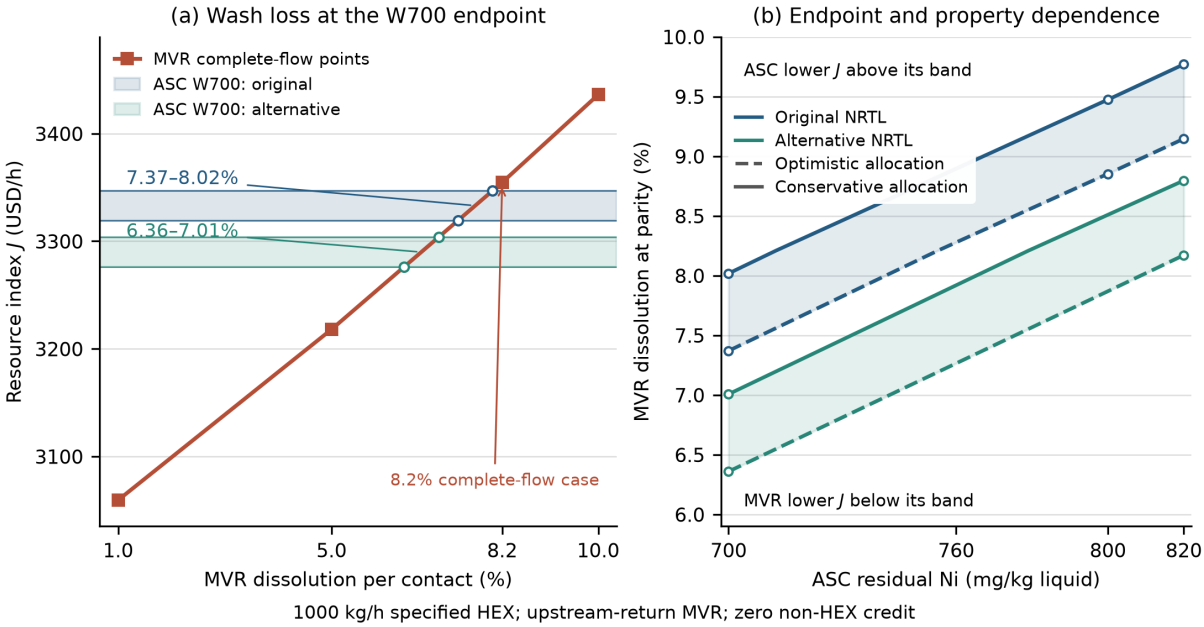


Figure S11: Conditional ASC–MVR resource-index boundaries at 1000 kg/h specified HEX and zero credit for retained non-HEX product Ni. (a) Squares are complete-flow MVR calculations; horizontal bands span the ASC W700 heat allocations for each acetone–water NRTL set. Open circles mark interpolated parity. (b) Piecewise interpolation combines original ASC endpoints at 700/800/820 mg/kg and alternative endpoints at 700/820 mg/kg with the MVR points at 1/5/8.2/10% dissolution. Dashed/solid curves use optimistic/conservative ASC heat allocations. The original aqueous MVR properties, upstream wash return and other route settings remain fixed. Bands compare heat allocations, and interpolation stays within the solved ranges.

Table S30: Retained nickel-bearing material in the wash comparison. All cases specify 1000 kg/h HEX. Non-HEX apparent salt has unresolved hydration and product acceptability; the main index gives it no credit.

Case	Non-HEX $NiSO_4$ / kg/h	Total product / kg/h
R95-W700	2.847	1003.475
MVR-WARM	2.878	1003.408
MVR-D050	13.085	1013.623
MVR-D082	20.031	1020.574
MVR-D100	23.547	1024.093

product Ni, giving 971.77 kg/h HEX and $J = 3259.9$ USD/h. These accounting changes retain the solved specific resource demands.

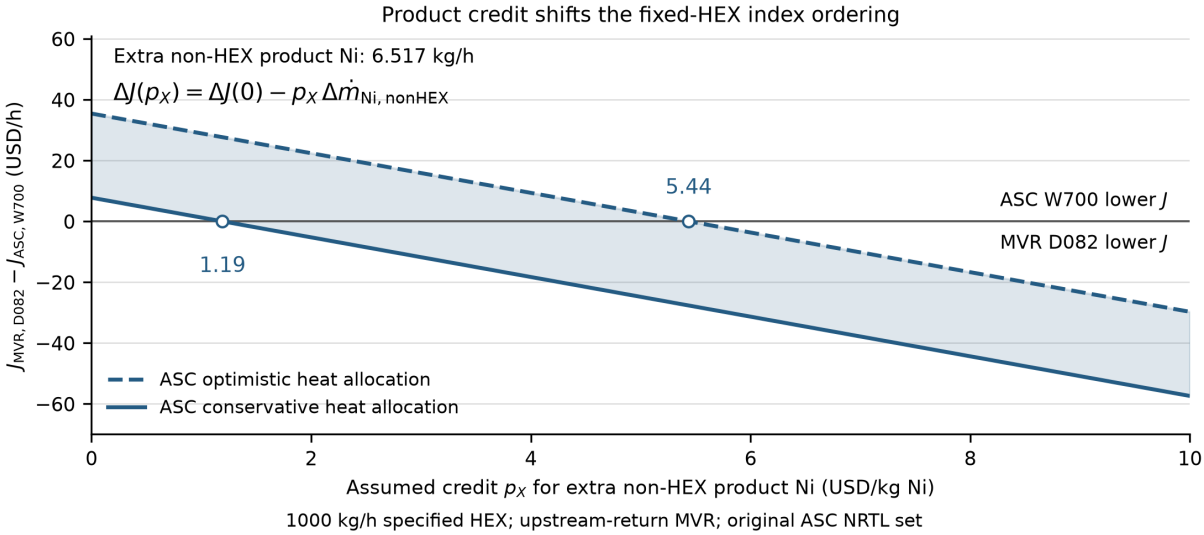


Figure S12: Product-credit sensitivity of the complete upstream-return 8.2%-dissolution MVR case relative to original-parameter ASC W700. At fixed 1000 kg/h specified HEX, p_X credits only the additional 6.517 kg/h non-HEX product Ni in MVR. The curves span the ASC heat allocations and cross zero at 1.19–5.44 USD/kg Ni. This algebraic sensitivity changes no process state; the retained material's final phase, quality and realizable value remain to be measured.

S10 Temperature profiles and heat-target calculation levels

The two vapor-recompression exchangers in the ASC W700 case contain extended phase-change plateaus (Fig. S13). Endpoint LMTD estimates are 871.2 and 208.1 m^2, whereas Eq. 3 gives 1898.9 and 484.2 m^2. Including P-FHX and W-HX gives 2542.3 m^2, compared with 1238.6 m^2 from endpoint estimates. At fixed states, $U = 350$, 700 and 1000 W m^{-2} K^{-1} gives approximately 5085, 2542 and 1780 m^2.

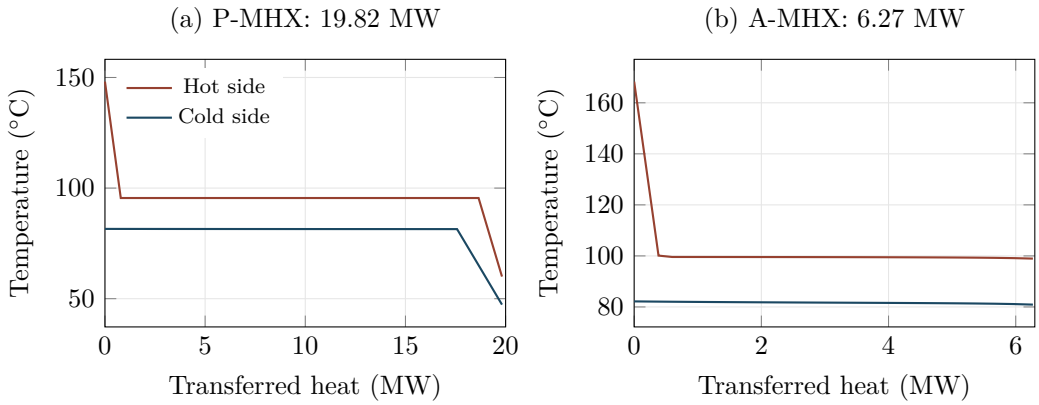


Figure S13: Countercurrent temperature–heat profiles of the two W700 MVR exchangers on the common heat coordinate used in the audit. Minimum sampled approaches are 12.68 and 17.54 K; segmented areas are 1898.9 and 484.2 m^2 at $U = 700$ W m^{-2} K^{-1}.

All four connected W700 exchangers satisfy the 10 K base approach: their minimum sampled

values are 17.54, 12.00, 12.68 and 11.31 K for A-MHX, P-FHX, P-MHX and W-HX. Three fall below 15 K. Accordingly, the ASC residual-network 15 K scan is not combined with MVR's 15 K result to rank redesigned plants. For the fixed MVR pressures, the 5/10/15 K hot targets are 0/0/32.57 MW and selected electricity is 2.373/2.395/4.156 MW; the last value includes both changed cooling allocation and lower refrigeration COP at evaporating levels of 10, 0 and −10°C. The large change identifies a heat-lift constraint. Increasing compressor discharge pressure could restore temperature driving force at higher compression work, requiring a new design calculation.

At 10 K, MVR recoverable-heat losses of 0/2/5% raise the hot target from 0 to 0.178 and 1.230 MW. Its final-stage inlet volume is about 595000 m^3/h in the saved model. Section S5 propagates specified vapor-pressure errors and suction losses through two-stage compression and the thermal cascade at the actual saved liquid composition. Its assumed error magnitudes are borrowed from pure-acid comparisons; they do not validate the high-acid ternary vapor pressure. The calculation identifies the consequence of an error without establishing that error's magnitude.

Constructive MVR heat matches. Table S31 supplies one explicit allocation of the saved MVR temperature–enthalpy profiles. Compressed bulk vapor releases 34.738 MW while cooling and condensing at 85°C/0.57776 bar; finisher vapor releases 6.217 MW with condensation at 35°C/0.056887 bar. The matched cold duties total 40.424 MW. Table S15 distinguishes the 58 mbar finisher compressor discharge from its 56.8871 mbar condensation endpoint. Bulk-vapor desuperheating serves product-water evaporation before the remaining superheat/latent heat serves E-BULK; the finisher service supplies evaporation/crystallization and the small component-cut heat balance. The segmented matches preserve at least 10 K everywhere on the specified profiles.

Table S31: Constructive additional MVR heat matches at $U = 700$ W m^{-2} K^{-1}. Areas use local countercurrent temperatures. The existing FEED-HX and final coolers are excluded. The last duty rounds to zero at three decimals (actual 0.0000413 MW).

Hot service to cold service	Duty / MW	Min. ΔT / K	Area / m^2
B-HX → E-BULK	34.542	10.00	4636.0
B-HX → D-EVAP	0.135	88.92	2.1
F-HX → EV-CRYST	5.746	10.00	819.5
F-HX → DESOLV trim	0.000	18.11	0.0

The additional area sums to approximately 5458 m^2, or 10915/3820 m^2 at $U = 350/1000$ W m^{-2} K^{-1}. The remaining cooling allocation is 1.210 MW cooling water plus 0.0650, 0.00102 and 0.13696 MW at the three refrigeration levels. It reproduces the original zero-heat cascade target and selected electricity. The construction permits staged exchange along each saved thermal profile, integrating each interval on its own phase segment so that temperature jumps are preserved at segment boundaries. Its 5458 m^2 covers the listed additional MVR matches; the 2542 m^2 ASC value covers already-connected exchangers. Their different service coverage is retained in equipment comparisons.

The changed MVR acid and return-location cases also admit explicit segmented matches (Table S32). Their fresh, composition-specific thermal profiles give zero external heat at a minimum local approach of 10 K. The remaining cooling duties and electricity reproduce each case's heat-cascade result to within 10^{-6} MW.

The 595000 m^3/h finisher suction corresponds to about 165 m^3/s. Dividing it among two or four parallel trains gives approximately 82.6 or 41.3 m^3/s per train. At an illustrative 25 m/s duct velocity, the corresponding circular inlet diameters are 2.05 and 1.45 m. Compressor selection must match these flow rates at roughly 20 mbar suction and the allocated suction-pressure loss. Air leakage and non-condensable balances determine startup and operating vacuum-service

Table S32: Additional MVR heat matches for the acid and return-location cases. All use $U = 700$ W m^{-2} K^{-1} and exclude existing FEED-HX and final coolers. Electricity includes compressor, modeled pump and allocated refrigeration demand.

Case	Matched heat / MW	Area / m^2	Electricity / MW
Acid 0.05 mol/L	36.572	5012	2.147
Acid 0.15 mol/L	45.328	5998	2.729
Downstream return, 1%	40.329	5446	2.400
Downstream return, 8.2%	41.581	5618	2.471

capacity.

S10.1 Feed preheating and outlet cooling

The sampled enthalpy curves identify a useful low-temperature heat sink in the solvent-recovery feed. In W700, RECOV condensation can heat PUMPED from 25.03 to 47.75°C with 2.117 MW; the purge then supplies 0.892 MW to reach 57.34°C. Both matches retain a 10 K minimum approach, and the sampled feed remains liquid. The corresponding P75 duties total 1.406 MW (Table S33). At $U = 700$ W/(m^2 K), the two added matches require approximately 213 and 99 m^2 for W700 and P75, respectively.

Sampling the two condenser paths also resolves most of the reference heat-allocation interval. W700 gives $J = 3346.67$–3346.73 USD/h and P75 gives 2893.59–2893.66 USD/h before changing the outlet boundary. Including purge cooling to 25°C gives 3344.32–3344.38 and 2892.27–2892.35 USD/h, respectively. These targets allow the newly included purge sensible heat to serve existing cold duties and assign the remaining cooling to the available utility temperature levels.

Reallocating the specified preheat duty from the reboiler to the feed lowers these common-outlet targets by 88.80 USD/h for W700 and 41.51 USD/h for P75. The resulting indices are 3255.51–3255.58 and 2850.77–2850.84 USD/h. Thus W700 has the larger additional heat-recovery opportunity, while P75 retains an approximately 404.7 USD/h advantage in this thermal screen. The composition-specific condenser and preheat calculations evaluate the 700 mg/kg design point. The endpoint sensitivity in the main text uses a separately stated constant-saving assumption.

Table S33: Two-stage feed-preheat targets from actual-composition enthalpy curves. Purge cooling is included to 25°C on both ASC configurations. Column product states and condenser duties are fixed in this screen.

Quantity	W700	P75-WASH
Condenser-to-feed duty (MW)	2.117	0.984
Purge-to-feed duty (MW)	0.892	0.422
Total feed-preheat duty (MW)	3.008	1.406
Feed temperature after first stage (°C)	47.75	47.68
Feed temperature after second stage (°C)	57.34	57.42
Total purge cooling to 25°C (MW)	1.559	0.730
Added match area at $U = 700$ (m^2)	213	99
Index saving on the common outlet basis (USD/h)	88.80	41.51

The P75 endpoint-price calculation uses the conservative endpoint allocation for all three complete states. At 700 mg/kg, the composition-specific condenser result is within 0.20 USD/h of that allocation; profiles at 1100 and 1400 mg/kg have not been evaluated. Favorable endpoint placement gives 1296 mg/kg parity, compared with 1226 mg/kg under conservative placement. These are heat-allocation alternatives. The separate preheat sensitivity subtracts 41.50768 USD/h at each endpoint, the incremental saving from the common-outlet to preheated common-outlet

Table S34: Heat-accounting levels for the same saved 700 mg/kg material states. Indices are USD/h at reference prices. Purge cooling changes the heat-network outlet boundary; preheat reallocates equal duty from reboiling to feed heating.

Calculation level	W700	P75
Endpoint heat placement	3319.20–3346.86	2874.43–2893.79
Composition-specific condensation	3346.67–3346.73	2893.59–2893.66
Purge cooled to 25°C	3344.32–3344.38	2892.27–2892.35
Fixed-product preheat target	3255.51–3255.58	2850.77–2850.84

target at 700 mg/kg. No revised endpoint composition, recovery-column traffic or recycling state is assigned by this shift.

Table S35: P75 endpoint requirements from saved-state price post-processing. The second column uses conservative endpoint heat allocation; the third additionally assumes a constant 41.51 USD/h saving. Values above the solved range are left unassigned.

p_N (USD/kg)	Nominal (mg/kg)	Preheat sensitivity (mg/kg)
6.2	702	926
8	1005	1183
10	1226	1369
12	1379	> 1400
13	> 1400	> 1400

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